Displaced Cuban Elites:

Local Economic Outcomes and Business Formation

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Abstract

How does a highly-skilled and entrepreneurial population, forced to move and without its capital, affect the places they arrive to? We study Cuba's Golden Exiles, the earliest wave of refugees arriving in the US after the 1959 Revolution. We identify the exiles by name, and link 1958 Cuban elite registers to Florida telephone directories and corporate filings. We study entrepreneurial activity and labor-market outcomes in neighborhoods where these exiles settled. Synthetic control estimates show elite settlement had a positive and significant effect of 2.20 times the pre-treatment average on business creation. Using matching estimators, we find that treated census tracts would exhibit a lower change in unemployment than comparable tracts. This is the first study to provide causal estimates of the effect of the Golden Exiles on the economy of Florida.

Keywords: Cuban Revolution, entrepreneurship, migration, human capital, enclave economy, business formation, Cuban exiles

JEL Codes: J61, L26, N46, N86, O15

1 Introduction

In October 1960, Ernesto Guevara, then president of Cuba's National Bank, told Julio Lobo, the island's leading sugar magnate, that the state would take his mills, and offered him a position running the nationalized sugar industry[1]. The offer showed that the new government valued not only what Lobo owned but also what he knew. Lobo asked for time to consider it and left Cuba within days, never to return (Thomas, 1977, pp. 520–521). The state kept his mills, and his knowledge of how to run them left the island with him. Thousands of owners and managers made the same choice. Between 1959 and 1964, roughly 270,000 Cubans, about 4 percent of the island's population, emigrated to the United States as the revolutionary government expropriated the firms, land, and financial assets of the business and professional class.[2] Business owners, managers, and professionals were heavily overrepresented among these emigrants (Fagen and Brody, 1964), who became known as the "Golden Exiles."[3] Those who left after 1960 arrived without their assets, carrying only their training, experience, and business networks. This paper studies how the settlement of this displaced business elite affected entrepreneurial activity and labor-market outcomes in the neighborhoods where they settled in Florida.

A large literature shows that skilled migrants raise innovation and entrepreneurship in receiving economies. Skilled immigrants contribute disproportionately to patenting (Hunt and Gauthier-Loiselle, 2010; Kerr and Lincoln, 2010; Akcigit et al., 2017) and to the founding of new firms (Kerr and Kerr, 2020; Azoulay et al., 2022), and easier access to skilled foreign workers increases firm size, productivity, and entry (Beerli et al., 2021). Forced migrants, however, differ from voluntary migrants in what they leave behind. They typically lose their assets and have little control over their destination (Becker and Ferrara, 2019). Research on

[1] Better known as "Che" Guevara.

[2] The population share uses the 1958 population of 6.95 million reported by United Nations (2024).

[3] Portes (1969, p. 508) describes the exodus as a "golden exile" because few immigrant groups had arrived in the United States so well prepared in education and occupation, and few had been received so favorably. That reception included admission as refugees and federal assistance with cash aid, training, and relocation (Mitchell, 1962).

forced migration has therefore focused on whether refugee inflows displace native workers, and most estimated effects on native employment and wages are small or statistically insignificant (Hunt, 1992; Friedberg, 2001).[4] Much less is known about whether forced migrants create firms in the places that receive them.

Several studies follow persecuted elites across borders to the places where they settled. Hornung (2014) finds that Prussian textile manufactories in towns with more Huguenot settlers were more productive more than a century after the settlers arrived. Moser et al. (2014) find that U.S. patenting rose by 31 percent in the research fields of German Jewish émigré chemists, mainly through the entry of new researchers into those fields. Toews and Vézina (2025) show that areas around Gulag camps that held more of the Soviet educated elite have higher wages, profits, and night-light intensity today.[5] In each case, the migrants carried scientific or technical knowledge whose returns depend little on personal wealth.

More closely, our study sits within two strands of the literature: how forced migration can push people into new occupations, and how these migrants affect entrepreneurial activity in their destinations. Forced migration moved displaced farmers out of agriculture and raised their long-run incomes in postwar Germany (Bauer et al., 2013) and Finland (Sarvimäki et al., 2022).[6] The Cuban exiles faced a similar break with their former occupations, since many arrived without assets, expected a short exile, and could not practice their professions without new licenses (Thomas, 1967, pp. 47–48, 55). Reviewing 59 studies of 19 displacement episodes, Verme and Schuettler (2021) find that research has concentrated on wages, employment, and prices over short and medium horizons, and that very few studies examine the production side of host economies or the entry and exit of firms. On this, Hessami et al. (2025) find that refugees arriving in German districts led to a 7.9 percent increase in business registrations,

[4] Becker and Ferrara (2019), Verme and Schuettler (2021), and Becker (2022) review this literature.

[5] Displaced political elites have also shaped mobilization and voting where they settled (Dippel and Heblich, 2021; Ochsner and Roesel, 2020).

[6] Displacement also raised the education and earnings of displaced children in Iceland (Nakamura et al., 2022) and shifted expelled Polish families toward education (Becker et al., 2020). Verme and Schuettler (2021) note that little is known about which skills the displaced bring and which occupations they enter in the host economy.

Akgündüz et al. (2018) and Altındağ et al. (2020) find that Syrian refugee inflows in Turkey lead to an increase in firm formation, and Jahn and Steinhardt (2025) find that the arrival of ethnic German migrants from Eastern Europe and the Soviet Union led to a net increase in business formation in Germany.

Research on Cuban migration has concentrated on the wage effects of the 1980 Mariel boatlift, whose migrants were less skilled than the earlier emigrants (Bach, 1980), with estimates ranging from negligible (Card, 1990; Peri and Yasenov, 2019) to substantial (Borjas, 2017).[7] The first wave has been studied mainly by sociologists of the Miami enclave economy,[8] and economists have paid it little attention, in part because its members cannot be identified individually in standard U.S. data sources. We provide, to our knowledge, the first estimates of the effects of this wave on the receiving economy.

We extend these strands of literature by going back to the earliest wave of Cuban exiles after the Revolution, and study the effects of this forced migration on the neighborhoods they settled in. The human capital we study is managerial and commercial, and putting it to use requires financial capital, credit, and commercial networks. Because the Cuban expropriations took away emigrants' capital, the episode allows us to study whether elite human capital persists once it is separated from any type of wealth that could have been carried over.

We identify the displaced before they left.[9] We digitize the 1958 edition of the *Libro de Oro de la Sociedad Habanera* (Álvarez de Cañas et al., 1958), Havana's social register, which lists 5,354 elite households and about 33,500 individuals with their addresses, and *Los propietarios de Cuba* (Jiménez Soler, 2008), which profiles 550 of the island's largest

[7] Clemens and Hunt (2019) attribute the disagreement to small survey subsamples. See also Bodvarsson et al. (2008) and Yao et al. (2022).

[8] See Wilson and Portes (1980), Wilson and Martin (1982), Portes (1987), Cobas et al. (1993), Portes and Zhou (1996), and Portes and Puhrmann (2015).

[9] Records of this kind are rare in historical settings (Becker, 2022), and existing studies assemble them from residence lists, passenger manifests, and the careers of dismissed academics (Buggle et al., 2023; Blum and Rei, 2018; Becker et al., 2024). These studies also show that emigration under persecution is selective, with selection strongest among the earliest emigrants (Blum and Rei, 2018). Cuban emigration shows the same gradient, as the professional and managerial share of arrivals fell from 31 percent at the onset of the exodus to 12 percent by 1970 (Pedraza-Bailey, 1985, pp. 11, 18).

business owners in 1958. To our knowledge, neither source has previously been digitized at the individual level or linked to records in a destination country. We link these individuals to Florida telephone directories from 1959 onward (Library of Congress, 2025), which yields 1,101 matched individuals, 710 of them geocoded, and to officer records in the Florida corporate registry (Florida Department of State, Division of Corporations, 2026), which yields 1,479 corporations created from 1959 to 1980.

We estimate the effect of elite settlement on entrepreneurial activity and labor-market outcomes. First, using the synthetic control method, we find that neighborhoods where elites settled saw increased entrepreneurial activity in the years after the exiles arrived. We argue this is because these exiles were a selected group of individuals, and the conditions they faced upon arrival made it more attractive to start a business than to enter the market for salaried jobs. Then, using matching estimators, we find that by 1980 treated neighborhoods had a lower change in unemployment than comparable controls, by an estimated 1.9 to 2.3 percentage points. This is consistent with the observation that Cubans entering the enclave in later years arrived in a welcoming economy that offered them better opportunities than those available in the first years after the Revolution.

Section 2 describes the expropriations and the emigration, and Section 3 sets out the conceptual framework. Section 4 constructs the roster, Section 5 describes the incorporation and census data, and Section 6 presents the two designs. Section 7 reports the results, Section 8 the robustness checks, and Section 9 concludes.

2 The Cuban Revolution and the Golden Exiles

Cuba in the late 1950s was a middle-income economy. In the 1955 benchmark comparisons of Devereux (2021, pp. 18–22), Cuban output per head is 27 percent of the United States level and consumption per head is 35 percent, a consumption level equal to that of Ireland and above those of Spain, Portugal, Italy, and Greece. Among the Latin American economies in

that comparison, only Uruguay, Argentina, Puerto Rico, and Venezuela had higher output per head. The 1951 World Bank mission to the island, known as the Truslow Report, described Cubans as "intelligent, able and quick to absorb modern knowledge" and Cuban businessmen as "shrewd and capable" (International Bank for Reconstruction and Development, 1951). The individuals we follow to Florida owned and managed the firms of that economy. In 1965 Fidel Castro acknowledged that managerial and technical knowledge on the island had been concentrated among its owners (Castro Ruz, 1965).[10]

Within a few years the revolutionary government replaced private ownership and market allocation with state ownership and central planning (Domínguez, 1978; Bastos et al., 2026). The Agrarian Reform Law of 1959 expropriated the large agricultural estates, and measures adopted in 1960 transferred foreign-owned oil refineries, U.S.-owned sugar mills, banks, utilities, and most large domestic industry to the state (Mesa-Lago, 1981; Jales et al., 2018). The decisive step for the domestic business class came in October 1960. After the United States banned most exports to Cuba on 13 October, the National Institute of Agrarian Reform took over 382 large private enterprises on 14 and 15 October, including all banks except two Canadian institutions, all remaining privately owned sugar mills, and a large number of distilleries, textile mills, rice mills, cinemas, and large retail stores. A second Urban Reform law issued at the same time limited each owner to a single residence, and another 166 U.S.-owned enterprises were nationalized on 25 October (Thomas, 1977, p. 519). The government subsequently extended central planning through JUCEPLAN, nationalized private education, converted cooperatives into state farms, and in 1963 implemented a second Agrarian Reform that reached medium-sized holdings. By 1961 the state owned banking, wholesale and foreign trade, and education outright, along with most of industry and transportation, and by 1968 industry, construction, and retail trade were also fully state-owned (Mesa-Lago, 1981, p. 15). The final stage reached the smallest proprietors. In the March 1968 *Ofensiva Revolucionaria*,

[10] In his speech of 19 February 1965, Castro said that the economy had been the Revolution's hardest front because "those who knew how to manage, those who knew the little technical know-how there was here, were the owners, the rich, all those people" (our translation).

approximately 56,000 remaining nonagricultural private businesses, including restaurants, repair shops, small retailers, and street vendors, were transferred to the state (Mesa-Lago, 1981, p. 23).

The extent of this transformation was not widely expected. Latin American revolutions were generally short-lived, and the director of the U.S. Cuban Refugee Program would later describe the stability of the Cuban revolutionary government as unexpected (Thomas, 1967, p. 48). Castro first described the Revolution publicly as socialist on 16 April 1961 and declared himself a Marxist-Leninist on 2 December 1961 (Thomas, 1977, pp. 534, 595). With the exception of a limited early emigration of government officials, bankers, and industrialists, most professionals and intellectuals remained in Cuba until the political direction of the regime became clear, and large-scale emigration began in 1961 (Wenk, 1968, p. 38).

The expropriated property belonged to identifiable individuals, and the earliest emigrants were drawn from among them. They came from Cuba's propertied and professional classes and included firm owners, executives, merchants, sugar-mill owners, manufacturers, cattlemen, and established professionals (Pedraza-Bailey, 1985; García, 1996). Emigration was a household decision made under expropriation and political pressure, and the order of departure followed political affiliation. Supporters of the Batista government emigrated immediately after its fall, and approximately 26,000 Cubans emigrated in the second half of 1959. Conservative opponents of the new regime followed in 1960 and 1961, when total emigration was approximately 110,000, and most liberal opponents emigrated between 1962 and 1964, when it was approximately 137,000 (Forment, 1989, p. 54). The occupational composition of departures changed in a similar sequence. The group associated with the Batista government numbered only a few thousand. As the state took over private property and enterprises, emigration of professionals and technicians increased, and by mid-1961 small merchants, clerical employees, and skilled and semiskilled workers were also emigrating in large numbers (Thomas, 1967, p. 47). By the late 1960s approximately 380,000 people, about one in twenty-one residents of Cuba, had emigrated (Portes, 1969, p. 505), and by 1972 the

cumulative total approached 560,000 (Bender, 1973, p. 271).

Until October 1962, emigration took place mainly on commercial flights between Havana and Miami. The United States closed its consulates in Havana in January 1961 but continued to admit Cubans under visa waivers. Arrivals peaked in 1962 at more than 1,800 refugees per week. When Cuba suspended commercial flights in October 1962, following the U.S. quarantine during the missile crisis, approximately 155,000 refugees had registered with the federal refugee center in Miami (Thomas, 1967, p. 47).

Two tabulations document the selection of these emigrants. Using records of the Cuban Refugee Emergency Center, Fagen and Brody (1964) compare refugees who arrived between 1959 and September 1962 with the 1953 Cuban Census, the last complete enumeration of the Cuban occupational distribution before the Revolution. Professionals and semiprofessionals were more than five times as common among refugees as in the Cuban workforce, and 14.2 percent of male refugees had completed four or more years of college, compared with 1.2 percent of Cuban adult men (Fagen and Brody, 1964, pp. 391–392).[11] A later tabulation by the same research group, reported in Pérez (1986, p. 129), finds that up to 37 percent of household heads arriving between 1960 and 1962 had been proprietors, managers, or professionals, compared with fewer than 10 percent of the Cuban labor force.[12]

Selection declined as the emigration continued. The professional and managerial share of arrivals fell from 31 percent at the onset of the exodus to 12 percent by 1970, when most arrivals held blue-collar, service, or agricultural occupations (Pedraza-Bailey, 1985, pp. 11, 18). Cuban authorities also began to screen departures. On 28 September 1965 Castro announced that any Cuban who wished to leave could do so (Bender, 1973, p. 271). The agreement that followed the resulting Camarioca boatlift established the Varadero airlift

[11] This covered 84,578 individuals classified by occupation and month of entry. Fagen and Brody (1964, p. 390) excludes students, housewives, retired persons without an occupational classification, and all individuals who entered before the change of government, which leaves 59,682. They mention that these numbers likely under-represent the earliest arrivals and wealthier refugees, as the Refugee Center did not open until early 1961.

[12] Fagen and Brody (1964) classify individual refugees by their own occupation, while Pérez (1986) reports the occupation of household heads, which would mechanically raise the professional and managerial share. However, the two agree on the direction and approximate magnitude of the selection.

and prohibited the emigration of men of military age and of professional, technical, and skilled workers whose departure would cause "a serious disturbance" in production or social services (Pedraza-Bailey, 1985, p. 16). The Freedom Flights that began in December 1965 brought approximately 260,500 Cubans by 1973 (Pérez, 1986, p. 130), with a less selected population. Our treatment window closes in 1964, so matched households belong only to the early and most selected cohort, taken from registers compiled before the Revolution. The Mariel boatlift of 1980 would then bring more than 123,000 people whose occupational profile differed significantly from that of the early emigrants (Bach, 1980, p. 39) and whose labor-market outcomes in Miami were considerably worse (Portes and Stepick, 1985; Portes and Puhrmann, 2015).

The assets that emigrants could transfer depended on the date of departure: restrictions were initially limited, and many emigrants who left in 1959 and 1960 were able to transfer a substantial share of their wealth. Toward the end of 1960, in the same months in which the state took over the remaining large Cuban-owned enterprises, the government began to confiscate the property of departing refugees, who thereafter arrived without assets (Thomas, 1967, p. 47). Restrictions on the removal of money and goods became fully binding in mid-1961 (Fagen and Brody, 1964, p. 390). Supporters of the Batista government, who left before the nationalization of the banks and the prohibition of currency exports, used personal savings and, in some cases, public funds to establish small businesses in Miami (Forment, 1989, p. 60). Professional and managerial households that emigrated from 1961 onward therefore transferred human capital and reputation, with little financial capital.

No established Cuban community existed to absorb the emigrants. Before 1959 the United States had no large Cuban settlement outside Florida (Thomas, 1967, pp. 51–52), and no more than 70,000 Cuban-born persons lived in the country (Pérez, 1986, p. 127). The nineteenth-century Cuban communities had formed in Tampa, Key West, and New York (Pérez, 1986, p. 126). Portes (1987, p. 340) describes Miami at the time as "a sleepy resort town at the southern tip of the United States," and studies of the following decades

describe its labor market as dominated by tourism and small industry (Wilson and Portes, 1980, p. 307). Its small Cuban population was broadly sympathetic to the revolutionary government (Forment, 1989, p. 60). The reasons emigrants reported for settling in Miami were largely unrelated to its commercial prospects: proximity to Cuba, a familiar climate, the prevalence of Spanish, and exile political organizations that sustained expectations of an early return (Thomas, 1967, p. 48).

The United States admitted Cuban emigrants as parolees without restrictions on residence or employment and did not establish refugee camps (Thomas, 1967, p. 48). Dispersal therefore relied on incentives.[13] In response to pressure on public services in Dade County, the federal government made resettlement outside Miami the principal objective of the Cuban Refugee Program, established in 1961 (Mitchell, 1962; Portes, 1969). Voluntary agencies matched refugee families with sponsors in other cities, and from late 1962 refugees who declined a reasonable resettlement offer lost financial assistance (Thomas, 1967, pp. 50–51). Between January 1961 and December 1966 the program resettled 134,945 of the 228,805 refugees it registered, distributing them across nearly 3,000 communities, with every state receiving some, for an average of fewer than fifty persons per community (Thomas, 1967, p. 51). Refugees who declined resettlement cited the climate, the proximity to Cuba, and the expectation of an early return (Thomas, 1967, p. 51).

The Cuban population of Miami nevertheless continued to grow. More than 100,000 Cubans lived in the Little Havana district in the early 1960s, and by 1970 more than half of the Cubans in the metropolitan area (56 percent) lived in the central city, compared with 26 percent of the total population (Aguirre et al., 1980, pp. 39, 46). Higher-income Cuban households later moved to Westchester and Kendall, in the southwestern part of the county (Portes, 1987, p. 349). The concentration of Cubans in Miami therefore reflects the location choices of the emigrants, made despite a federal program that encouraged and financed

[13]Other host governments have assigned forced migrants to locations directly. In postwar Germany, occupation authorities placed expellees in rural districts with available housing (Peters, 2022), which limited their choice of location (Braun and Mahmoud, 2014), and the assigned locations affected their subsequent integration (Braun and Dwenger, 2020).

settlement elsewhere.

Cuban-owned businesses appeared within a few years of the first arrivals. By 1966 more than 3,000 businesses in Miami operated under Cuban management, and Cubans had also established a small number of sugar refineries and tobacco factories (Thomas, 1967, p. 55). Firms founded by emigrants could in turn employ later Cuban arrivals. Wilson and Portes (1980) find that Cubans employed in Cuban-owned firms obtained returns to education and occupational experience comparable to those of non-immigrant workers outside the enclave, contrary to the prediction that immigrants are confined to a low-wage secondary segment.[14]

3 Conceptual Framework

In this section we explain why we expect the Golden Exiles to affect the neighborhoods where they settled. We set out the mechanisms through which we expect entrepreneurial activity to increase, and the effects this would later have on the Cuban enclave.

First, Locay and Sanguinetty (1996) suggest why the Golden Exiles might be disproportionately composed of people with entrepreneurial skills through selection into migration alone. They argue this for two reasons. First, entrepreneurs had the most to lose from the Revolution's confiscation of assets, losing both their assets and their livelihood. Second, entrepreneurs tend to be more willing than the average person to take risks, which in itself makes them more likely to take the risk of migrating and starting over in a new country. Selection would therefore leave Cuban migrants, especially those who left immediately after the Revolution, with a disproportionate share of entrepreneurial skills. If these skills are transferable, selection into migration is one mechanism through which we would expect a higher degree of business creation in the places where they settle.

We also take the dual labor market hypothesis from Doeringer and Piore (1971) as a reason to expect higher entrepreneurial activity among exiles. This theory posits a primary sector

[14]The advantage of enclave employment did not extend to later arrivals, and Portes and Puhrmann (2015) document that pre-Mariel cohorts retained it while later cohorts were largely excluded.

with stable jobs, returns to education, and internal mobility, and a secondary sector with low wages and limited mobility, to which immigrants are typically expected to be confined. Professional licenses were not transferable, and only about one quarter of registered refugees spoke English well enough to be employed (Mitchell, 1962, p. 5). Many professionals retrained for other occupations, with former lawyers employed as teachers, librarians, and insurance adjusters (Thomas, 1967, p. 54). Barriers to resuming prior occupations, together with facing the secondary sector, would result in a "downgrade", which would make entrepreneurship a more attractive option for the Golden Exiles.

In addition, once the exiles realized that their stay was not temporary and that they would not be able to return, they could commit to longer-term projects. If the chance of returning to the island was gone, then building a business would not carry the potential sunk costs of returning and leaving it behind. Finally, the family and business networks formed in Havana eventually moved with them, and networks of this kind gave immigrant entrepreneurs access to the start-up capital, customers, and information that new firms require (Portes, 1987, p. 346). For all these reasons, these individuals, who by selection alone would have more experience using their alertness to profit opportunities (Kirzner, 1973), would meet conditions that would push them towards using it when they arrived in the US. We expect a higher rate of entrepreneurial activity in the neighborhoods where the exiles settled.

The Cuban enclave would form around the neighborhoods where these exiles settled and would receive thousands more migrants. A higher degree of entrepreneurial activity could benefit successive immigrant waves by offering them opportunities upon arrival, which would also allow them to access higher wages than comparable migrants. As reported by Wilson and Portes (1980), this would be the case: Cuban migrants would arrive at an enclave that welcomed them with primary-sector opportunities, unlike earlier migrants. We argue that an increase in entrepreneurial activity would allow Cuban enclave neighborhoods to achieve labor-market outcomes similar to those of comparable non-migrant neighborhoods. Golden Exile settlement would prevent the Cuban enclave from exhibiting higher unemployment rates

and would keep wages and income from falling below those of comparable neighborhoods.

4 Matching Cuban Elites to Florida Landings

In this section, we describe how we obtain the names of the pre-revolutionary Cuban elite and how we match them to U.S. records to locate where they settled in Florida.

4.1 The 1958 Cuban elite registers

The *Libro de Oro de la Sociedad Habanera* (Álvarez de Cañas et al., 1958) is an annual social register of Havana. Its editors compiled each edition as an alphabetical list of families, and its unit of observation is the household. The 1958 edition lists the fifty noble titles then current on the island. The register also carries short biographies of prominent physicians, lawyers, and teachers, and lists the couturiers who served this clientele together with the addresses of their ateliers. Its coverage overlaps closely with the private clubs that organized elite social life, and Roorda (2004) finds roughly four fifths of the Havana Country Club's Cuban membership listed in it.

Each record is built around a household head and spouse and then lists their children and their children's spouses. It gives the home address, telephone number, and neighborhood, and in most cases the head's profession and work address. The addresses are precise enough to place families street by street, and Vázquez Fleitas (2024) uses those in the 1953 edition to map the elite across Havana's neighborhoods.

Three features of the register suit it to our purpose. It lists professionals, executives, and owners of capital, which is the human capital we want to follow. Because it centers on Havana, it covers the city where Cuba's commercial, financial, and professional class was concentrated. And because it appeared annually, each edition fixes the roster to a date, so the 1958 volume captures the elite as it stood on the eve of the Revolution.

Los propietarios de Cuba (Jiménez Soler, 2008) adds the economic dimension. It is a

biographical dictionary of the island's largest business proprietors as they stood in 1958. Each of its 550 entries records the proprietor's holdings, positions, family, last address and social clubs, nationality, profession, and the origin of the fortune, and rates the proprietor's economic importance on a five-point scale. The two sources describe overlapping populations, and together they cover both the social and the economic elite.

Historians of pre-revolutionary Cuba have used both works.[15] We use them as a sampling frame. Together they name the people whose departure we trace, and those names are what we carry into the United States records described next.[16]

4.1.1 White pages: locating the elite in Florida

To locate where elites lived after moving out of Cuba, we use the Library of Congress collection of digitized telephone directories (Library of Congress, 2025). Here, we were able to find directories from 1959 onward. The listing format for individuals looks like the following:

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Alvarez Enrique 20141 NW 15 Av 624-0296
Alvarez Argudin Mario 211 Antilla AvCG--445-3207
Alvarez Aramis Jr 3900 Gardn Av MB---JE 4-2055
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For matching, names are stripped of accents, honorifics and punctuation, then split into a first given name, a paternal surname and a maternal surname. A person recorded in Havana as "Márquez Sterling, Carlos" may appear in a U.S. source filed compound (*Márquez Sterling Carlos*) or as paternal-only (*Márquez Carlos*).

We are able to match 1,101 individuals, with 992 of them having a usable street address, and after geocoding them to get coordinates we get 710 landing geolocated. We screen these

[15] Roorda (2004) uses the *Libro de Oro* to map the membership of Havana's clubs and Guerra (2018) to identify individuals. González Acosta (2022) draws on the 1958 edition to trace surviving noble titles, and Vázquez and Pombo (2013) draws on the register to reconstruct the couture trade. Vázquez Fleitas (2024) tabulates the 1953 edition by neighborhood. Vázquez Fleitas (2023) draws on both the *Libro de Oro* and *Los propietarios de Cuba*, and reads the *Libro de Oro* for 1947 through 1958 against the *Registro Social de La Habana* for 1950 through 1958 to observe turnover in the roster. Pérez (2023) follows one family across the Revolution through the *Registro Social*.

[16] We hold editions of the *Libro de Oro* for 1947 through 1960. We use 1958 because it is the last edition compiled before the Revolution. The 1953 edition contains 5,791 families (Vázquez Fleitas, 2024), so the register is stable in scale across the decade. The full series would let entry and exit be observed year by year.

matches to lower the chance of them being spurious. If an individual is matched on both surnames, we accept it outright. If it is matched only on the paternal (first) surname, we accept it only if the surname does not appear more than twelve times in the directory volume. This guards us agains having matches with individuals that have very common surnames, like García or Rodríguez. 92% of matches are paternal-surname based, and only 8% use both surnames. We take each of these matches and locate them in a 1960 census tract.

Figure 1 shows where Golden Exiles established their residences when they arrived to Miami. Every point is an elite household, and the shading signals the 1960 Spanish-surname population share for each tract. Most of the households fall outside of what were the major Hispanic-populated areas.

Table 1 compares the occupations of the matched households with those of all households in the 1958 register. No occupational group differs from its share in the register by more than 2.5 percentage points. These linked households are the basis of the treatment definition in Section 6.

4.1.2 Florida corporate filings: locating their firms

We can also track where and when the exiles founded businesses using the Florida Division of Corporations (Sunbiz) bulk filing database (Florida Department of State, Division of Corporations, 2026). This dataset contains business entities registered in Florida, with filing dates going back to the nineteenth century. For each firm, the data record the corporation name, officer fields, filing date, status, and last-event date.

We match the same elite names against the Florida corporate registry with the intention to identify elite-owned firms. We read each entry's document number, entity name, filing year and principal address, as well as up to seven officer name slots per corporation. Here we apply a looser restriction to paternal-surname-only matches, since the registry contains many more names: we require the full given name and a surname that appears no more than twenty-five times among the registry's officers. We count a firm only once even if it has two

officer matched. This procedure yields 1,479 matched corporations.

Figure 2 shows the matched businesses by decade of first appearance. A small number were founded in the 1960s, concentrated along 8th Street (*Calle Ocho*) and West Flagler. Elite-linked businesses expanded in the following decade, with clusters in Little Havana, Coral Gables, Downtown, and Brickell. The yearly and cumulative counts of these filings are shown in Figure 3.

The name-matched firms are used only for the figures in this section, since treatment is defined by where the elite settled rather than where they founded firms. The business-formation outcome described in Section 5 is instead an aggregate count of all incorporations drawn from the same registry.

5 Data

Given our conceptual framework, we study two different types of outcomes: entrepreneurial activity and labor-market outcomes. We measure entrepreneurial activity as new business creation in the Florida corporate filings database, and we take labor-market outcomes from census data.

5.1 Florida Corporate Filings

Business formation comes from the Florida Division of Corporations bulk filing database (Florida Department of State, Division of Corporations, 2026). We parse 668,468 filings with filing years between 1940 and 1980. Of these, 398,740 carry a valid Florida zip code in the registry, and we recover a zip code for 57,875 more by geocoding their street address, for a total of 456,615, or 68.3 percent, spread across 1,896 distinct zips.[17]

[17]ZIP codes were introduced in 1963, so only firms that had activity in their filings after this date have a ZIP code in their address. On top of that, some firms from after 1963 do not have a ZIP code, making the count of firms per zip code lower than the real one. To solve this, we recovered – with the help of the US Census Geocoder – 57,875 from the 70,022 filings that have a full address but no ZIP. This implies an increase of 24% in pre-treatment filing records. There are a further 149,000 filings without any street address, which we are not able to recover.

This series is an aggregate count over the entire registry. It counts every incorporation in a zip-year, and it is different from the 1,479 elite-matched corporations of Section 4.1.2. Treatment is assigned based on settlement, not firm foundation.

5.2 Census Data

We use census data to analyze labor-market outcomes. We follow the outcomes studied in the Mariel literature as closely as we can: wages, unemployment, and employment. However, the CPS microdata used by Card (1990), Borjas (2017), and Peri and Yasenov (2019) are only available from 1962, which postdates the start of our treatment period. As a substitute, we use the National Historical Geographic Information System, which provides tract-level tabulations of the 1950, 1960, 1970, and 1980 censuses (Manson et al., 2026). We estimate effects on three outcomes: the unemployment rate, the employment-to-population ratio, and log median family income. No tract-level wage exists for 1960, so median family income is the closest analog we can find in this setting.

Tract boundaries change between censuses, so we harmonize every year to 1960 boundaries by apportioning each earlier or later tract's counts according to the share of its land area falling in each 1960 tract, following Lee and Lin (2018, p. 672).[18] The 1950 census is the binding constraint on tract coverage, as we include 1950 characteristics in the analysis. This also means that the only metropolitan area we will see in Florida is Dade county, as it was the only one tracted in 1950.

We draw controls from two pools, holding the treated set fixed: the rest of Dade County and the 26 comparable metropolitan counties – those with at least 20 tracts and 50,000 people in 1950 and similar population and population growth to Miami. Control tract vary as variable coverage changes depending on the outcome being observed. There are 47 control tracts in Dade and 2,234 in comparable metropolitan areas.

[18]We recompute rates and shares from the apportioned counts. For median variables, we calculate a weighted average across overlapping tracts using the intersection area multiplied by population as the weight. Only 1960 values are observed directly.

For matching, we use five 1960 characteristics in all estimations: log mediam family income, median school years, Spanish-surname share, professional and managerial share, and log population density. Additionally, we add the outcome's 1960 and 1950 levels. Table 2 shows the summary statistics for the matching covariates for treated tracts and the control pools.

6 Empirical Strategy

In this section, we explain how treatment is defined and the research designs applied to study the outcomes mentioned in the previous section.

6.1 Defining treatment

We define treatment using elite settlement. Elite individuals matched to the Florida telephone directories with a street address, from 1959 to 1964. The 1964 cutoff is set so we only capture the earlier wave of migrants, as the Freedom Flights would begin in 1965, bringing a much less selected inflow of migrants to the same places.

We locate each of the landings in census tracts and zip codes, setting five elite households as the minimum to establish treatment in a tract or zip code. For census tracts, which are only in the Dade area, we are able to locate 414 landings across 97 tracts, with 26 of them receiving at least five landings, and 23 surviving into the estimation sample. Then, for zip codes, since we have access to the whole state of Florida we are able to locate 504 landings across 79 zips. Of these 79, 30 receive at least five, and they are spread across Miami-Dade (26), Tampa (3), and Broward (1). These 30 zip codes received 419 of the 504 landings.

We set the threshold at five households, on the grounds that this is a relevant presence in the neighborhoods where exiles settle. The average size of a matched household is 6.97 people, which would mean that elites are between 0.5 and 1.8 percent of the population in treated tracts. Since we are eliminating households with common last names, we argue this

is a floor for how many of our exiles are actually settling in these places. This threshold represents a presence that could plausibly matter for a neighborhood. Then, we apply the same threshold to our business formation analysis using zip codes.

6.2 Synthetic control for business formation

We estimate the effect of elite settlement on business formation by synthetic control (Abadie et al., 2010). Each of the 30 treated zip codes – those receiving at least five elite households – is matched to a weighted combination of 196 donor zip codes, using annual counts from 1946 to 1980 with treatment beginning in 1965. Zip codes that receive one to four elite households are dropped rather than used as donors, following the same logic as the tract design. We choose 1965 as the treatment year, as this is our cutoff for elite arrivals into Florida: this is the year where Freedom Flights began and the Golden Exiles had already established themselves.

The outcome is each zip code's incorporations divided by its own 1946–64 mean. This means that the effect should be read as a multiple of its own pre-treatment average, and that every unit averages one before treatment. We normalize because many treated zip codes had pre-1965 incorporation levels above those of every donor. Normalizing removes these level differences, so the synthetic counterfactual tracks proportional changes rather than levels. In Figure 4 we plot the trajectories of treated and donor zips before and after doing the normalization.

The filings hold 1,817 Florida zip codes that receive no elite household. Fitting a synthetic control on a pool that size can lead to interpolation bias (Abadie, 2021, p. 401) due to many dissimilar units, and most of these zip codes cannot be used in any case: normalization divides each zip code by its own pre-treatment average, which rules out every zip code with no filings before 1965. We therefore restrict the pool on pre-treatment activity. Of the Florida zip codes that appear in the filings and receive no elite household, we keep the 196 with at least 38 filings between 1946 and 1964, an average of two a year. This restriction operates as a floor,

discarding zips that have too little business activity to be comparable to our treated zips. This matters because of normalization: a zip code with a very low pre-treatment average might seem like it "boomed" with a much smaller increase in the outcome than other zips. A zip averaging one firm per year before treatment that had twenty after treatment would read as an increase of twenty times, so if these are included in the estimation, they would bias our estimates towards zero or below.[19]

We use all nineteen pre-treatment years as predictors, following Ferman et al. (2020), as we do not have zip-level covariates for this period. Estimations are done using `allsynth` (Wiltshire, 2022): a separate fit for each treated zip code against a common 1965 treatment date, averaged across the 30 as in Cavallo et al. (2013), with the gap bias-corrected by ridge regression (Ben-Michael et al., 2021; Abadie and L'Hour, 2021) as in Bastos et al. (2026).

For inference we use placebo-based tests. We follow Abadie et al. (2010) by fitting a synthetic control for each of the 196 donors against the rest of the pool. Since we have 30 treated zips, we also follow Cavallo et al. (2013) by averaging bundles of 30 donors. We then take 100,000 bundles, with donors chosen at random with replacement to get our placebos. For the overall p-value, we use the share of bundles whose post- to pre-treatment RMSPE ratio is larger than the treated average's. For the per-year p-values, we divide each year's gap by its own pre-treatment RMSPE for the treated and every bundle, and use the share of bundles whose scaled gap is larger in absolute value.

6.3 Census outcomes: Matching estimators

For each outcome we ask how much more treated tracts changed between 1960 and the endpoint than tracts that resembled them beforehand. The outcome is therefore a change rather than a level, and the estimate is the contrast between that change in treated tracts and the same change in a weighted set of matched controls. The nine estimators described

[19] Including all donors with enough data – every zero-elite zip code with at least five filings between 1946 and 1964, 727 in total – leads to an estimated effect of +0.62, against +2.20 in the adopted specification, consistent with the bias described above.

below differ only in how they choose and weight those controls.

Matching balances the pre-treatment characteristics we observe for the census tracts, while differencing removes unobserved tract characteristics that do not change over time (An and Winship, 2017). We follow Grier and Grier (2021) and Bologna Pavlik and Young (2026) in matching treated units to controls and estimating the effect on a differenced outcome.[20]

We match on the outcome's own 1950 and 1960 levels together with five 1960 structural characteristics: log median family income, median school years, the Spanish-surname share, the professional and managerial share, and log population density. Two pre-treatment periods are the most the data allow. The census is decennial, and 1950 is the first year in which Miami was tracted at all, so no earlier tract observation exists for any of our units.

Additionally, we run a "placebo," which is the change between 1950 and 1960 for our outcomes, matching on 1950 characteristics. Matching on observables cannot fix selection on where elites chose to settle, and differencing only sweeps out unobservables that are fixed over time, not the neighborhood trajectories that could have driven the choice. The placebo checks that the treated tracts were not already on a different trajectory from those they are compared to. Since there was no treatment during this time, there should be no significant effect on treated tracts from elite settlement.

We report nine matching estimators: propensity-score matching at one to four nearest neighbors and with a normal kernel, and Mahalanobis matching at one through four nearest neighbors. Inference differs by family: we bootstrap the average effect on the treated with 250 replications for the propensity-score estimators, and use the bias adjustment of Abadie and Imbens (2011) with the robust variance of Abadie and Imbens (2006) for the Mahalanobis estimators.

[20]Our implementation follows the public replication code of Bologna Pavlik and Young (2026), using `psmatch2` (Leuven and Sianesi, 2003) and the Mahalanobis estimators with Stata's `teffects nnmatch`, but we follow Grier and Grier (2021) by using the option `common` so treated units that do not have any comparable controls are dropped in the PSM specifications. This removes between 12 and 17 of the 23 treated tracts when using the Dade pool, depending on the outcome and horizon, and none in the comparable-metro pool in the PSM estimations.

7 Results

In this section, we present the results, starting with entrepreneurial activity and then turning to labor-market outcomes.

7.1 Entrepreneurial activity

Our results for business formation are shown in Figure 5. Before treatment both the treated zip average and the synthetic control are relatively flat and follow each other's trajectory closely. Following treatment in 1965, the treated zips are indistinguishable from their synthetic control for three years. The effect turns positive in 1968 and is significant at the 10% level in nine of the sixteen post-treatment years – 1968 through 1971, 1973, 1974, and 1976 through 1978 – and at 5% in eight of them, 1977 being the exception. It is not significant in 1979 or 1980. Because the outcome has been normalized, the result is not in terms of newly formed businesses, but in terms of the pre-treatment average.

In the years with significant positive effects, the gap ranges from 0.88 to 4.59 times the pre-treatment average. Over the full post-1965 period the average effect is 2.20 times the pre-treatment average (Figure 6, panel b). Business formation in the zip codes where the elite settled rose relative to the synthetic counterfactual. The lag in these estimates is itself informative. Business formation does not rise until 1968, which is what would happen if the capital these migrants held had to be rebuilt before they could start businesses.

We also estimate a ladder to see how the effect changes as the treatment threshold rises. In Figure 6 we show the synthetic control gaps for a set of different thresholds starting at 3 landings per zip, through the adopted 5, up to 20 landings (which yields only seven treated units). As the figure suggests, the effect of elite landings on business formation rises with the treatment threshold, though not at every step. The respective ATTs and p-values are shown in the right panel, with the largest ATT being that of the largest dose (20 elite landings per zip).

The ATT of the 20-landings-per-zip specification is 3.96 times its own pre-treatment average, significant at the 5% level. This estimate rests on seven treated zip codes, which cover what became the core of the Cuban enclave in South Florida: Little Havana, Coral Gables, Brickell / east Little Havana, Little Havana west of downtown, Coral Way, Westchester, and Edgewater.

The results are consistent with the expectation that the Golden Exiles were a group inclined toward entrepreneurship, given their selection into migration, their reduced opportunities upon arrival, the no-return condition, and the transfer of their networks and contacts to Florida. We now turn to labor-market outcomes, to study how this greater propensity for entrepreneurship affected workers coming into the Cuban enclave.

7.2 Labor-market outcomes

In this set of results, we study unemployment, employment, and median family income. Table 3 summarizes the results and the full tables with all nine estimators are in Tables A1–A6 in the appendix. We report the Average Treatment Effect on the Treated on the change in the outcome variables from 1960 to 1970 and from 1960 to 1980, against two control pools: the rest of Dade County, and the 26 comparable metropolitan counties. The appendix tables also report the covariate balance χ^2 after matching.

The results for the change in unemployment can be seen in Table 3. First, from 1960 to 1970, we see that when compared to other Dade tracts the effect is significant in four out of nine estimators, with all estimates being positive, and when compared to tracts in comparable metros, there is a statistically significant negative effect on the treated tracts. Then, for the change from 1960 to 1980, both control pool specifications show a negative effect that is statistically significant in most estimators.

Over the first decade, the result therefore depends on the comparison. Treated tracts are not significantly different from Dade County tracts in a majority of estimators and they fare better in this measure than tracts in the comparable metros. By 1980 the change in

unemployment in treated tracts was lower than in matched tracts in both pools. We do not observe any significance in the placebos, indicating that treated tracts were not already on a different trajectory from controls.

These results are consistent with the Cuban enclave offering primary-sector outcomes to the Cuban migrants who joined it. In this case, we even see a lower change unemployment rate than in comparable census tracts, which gives some insight into how many opportunities were offered to later waves of migrants.

Then, we turn to employment and median family income. There is some statistical significance spread across time periods and control pools, but only the 1960 to 1980 change in median family income is significant and positive in the majority of estimators. However, the comparable-metro placebo tests are significant for both outcomes, which might indicate that the neighborhoods were already on a different trajectory. This would invalidate the positive effect we see in median family income, and could be interpreted as the treated tracts "bouncing back" from having a lower change in the pre-treatment years. Because of this, we do not take these two outcomes' results as informative.

8 Robustness

8.1 Synthetic DiD for business formation

As an alternative estimator, we use synthetic difference-in-differences, developed by Arkhangelsky et al. (2021), using the `sdid` Stata package by Clarke et al. (2023). Under this estimator, using the same treatment threshold as the original synth, and the exact same pool of donors, we see an ATT of 39.89 firms per zip per year (Table A7 and Figure A1), significant at the 1% level using bootstrap inference techniques. Because this outcome is measured in levels, the magnitude is not directly comparable to the normalized estimate, but the effect grows over the post-treatment period as it does in the main specification. The event-study estimates are positive and significant at the 5% level in every post-treatment year except 1972, rising from

about 5 firms per zip code in 1965 to about 99 in 1980. This pattern is consistent with the main estimate.

8.2 Moving treatment back to 1959

We also run the main synthetic control specification with the treatment date moved back to 1959, the year of the Revolution. This yields an ATT of 3.68 times the pre-treatment average. The years 1959 to 1964, which are pre-treatment in our main specification, show a negative but relatively small effect here, significant at the 10% level in 1962 and at the 5% level in 1964. These are the years in which the exiles were still settling in Florida, so a negative effect could be expected. The positive and significant effect still appears later in the post-treatment period.

8.3 Varying treatment threshold for labor-market outcomes

Since we see the effect of elite landings grow as the threshold increases for business formation, we also ran the labor-market outcomes at different treatment thresholds. We examine how the effect changes if the treatment threshold is varied between three and seven households per tract. We report a summary table (Table A8) which shows the proportion of the estimations that are significant, as well as the median estimated effect. The unemployment result holds under any treatment threshold, with the others remaining not significant, and some gaining and losing significance for employment and median family income.

9 Conclusion

The Cuban Revolution expropriated the firms, land, and financial assets of the island's business and professional class, and many of its members left for Florida between 1959 and 1964. Those who left after 1960 arrived with their training and business ties but little of their wealth. This separation of skill from capital lets us ask what a skilled population does to a

receiving economy when it arrives without the savings migrants usually bring.

There is an economic literature on Cuban migration, but it focuses mostly on the Mariel boatlift of 1980, a sudden shock to the receiving economy in South Florida that is well suited to studying the effects of migration. We study the earlier wave of migrants from Cuba to Florida after the Revolution and provide the first evidence of how this movement of people affected the places where it settled. Additionally, we add to the forced migration literature by providing a new case study, where a highly skilled group of immigrants arrived at a place where an economic and cultural enclave formed around them.

To study the Golden Exiles, we pull together prerevolutionary Cuban registries, early 1960 Florida telephone directories, Florida Corporate Filings, and census data. We observe the Golden Exiles' effect on business formation, seeing that neighborhoods where they settled experienced an increase in entrepreneurial activity. We do not see consistent differences in income or employment between the areas where exiles settled and the control areas, but we do find that treated areas experienced a more favorable change in unemployment than comparable controls.

The timing of these results fits the mechanisms we describe in Section 3. The early exiles were already a selected group of people with more entrepreneurial characteristics, and they carried the know-how of having run firms and practices in Havana along with a network of family and social ties that accompanied them into exile. Many could not return to their former occupations, since professional licenses did not transfer, and only about a quarter of arrivals spoke enough English to hold a job (Mitchell, 1962). Founding a firm required neither, which could push them into business once they had rebuilt some of the capital a firm needs. That delay is consistent with incorporations pulling ahead only in the late 1960s.

Then, firms founded by exiles could hire later Cuban arrivals, and provide them with returns to education and experience comparable to those of workers outside the enclave (Wilson and Portes, 1980). This is consistent with lower unemployment by 1980 where the elite settled. The Freedom Flights brought some 260,500 Cubans between 1965 and 1973

(Pérez, 1986, p. 130), and our outcomes count every incorporation and every resident, so the effects include the arrivals, customers, and firms that elite settlement drew.

Interpreting these results under the light of the sociological literature that has studied this episode, we provide evidence to confirm that the Golden Exiles were people capable of performing business, and that this would "set up" the success of the Cuban economic enclave later on. Human capital and entrepreneurial skill crossed the strait, and the conditions led exiles to exercise their alertness. We are the first to provide causal evidence of the effect of the early Cuban migration to South Florida.

Lastly, our results also inform research on the economic consequences of the Cuban Revolution, which has estimated its effects on GDP per capita and exports (Jales et al., 2018; Bastos et al., 2026), infant mortality (Geloso and Bologna Pavlik, 2021), and environmental performance (Martinez and Geloso, 2026), and has asked what Cuban living standards would have been without it (Ward and Devereux, 2012). Devereux (2021, pp. 12–13) attributes part of the 15 percent fall in Cuban GDP per capita between 1957 and 1963 to the departure of managers and skilled workers. Because these studies measure the combined effect of the Revolution, none of them isolates this channel. Our results address it from the receiving side. The human capital that left Cuba remained productive, and within about a decade it raised entrepreneurial activity where the emigrants settled.

References

Abadie, A. (2021). Using synthetic controls: Feasibility, data requirements, and methodological aspects. *Journal of Economic Literature*, 59(2):391–425.

Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.

Abadie, A. and Imbens, G. W. (2006). Large sample properties of matching estimators for average treatment effects. *Econometrica*, 74(1):235–267.

Abadie, A. and Imbens, G. W. (2011). Bias-corrected matching estimators for average treatment effects. *Journal of Business & Economic Statistics*, 29(1):1–11.

Abadie, A. and L'Hour, J. (2021). A penalized synthetic control estimator for disaggregated data. *Journal of the American Statistical Association*, 116(536):1817–1834.

Aguirre, B. E., Schwirian, K. P., and La Greca, A. J. (1980). The residential patterning of Latin American and other ethnic populations in metropolitan Miami. *Latin American Research Review*, 15(2):35–64.

Akcigit, U., Grigsby, J., and Nicholas, T. (2017). Immigration and the rise of American ingenuity. *American Economic Review*, 107(5):327–331.

Akgündüz, Y. E., van den Berg, M., and Hassink, W. (2018). The impact of the Syrian refugee crisis on firm entry and performance in Turkey. *World Bank Economic Review*, 32(1):19–40.

Altındağ, O., Bakış, O., and Rozo, S. V. (2020). Blessing or burden? Impacts of refugees on businesses and the informal economy. *Journal of Development Economics*, 146:102490.

An, W. and Winship, C. (2017). Causal inference in panel data with application to estimating race-of-interviewer effects in the general social survey. *Sociological Methods and Research*, 46:68–102.

Arkhangelsky, D., Athey, S., Hirshberg, D. A., Imbens, G. W., and Wager, S. (2021). Synthetic difference-in-differences. *American Economic Review*, 111(12):4088–4118.

Azoulay, P., Jones, B. F., Kim, J. D., and Miranda, J. (2022). Immigration and entrepreneurship in the united states. *American Economic Review: Insights*, 4(1):71–88.

Bach, R. L. (1980). The new Cuban immigrants: Their background and prospects. *Monthly Labor Review*, 103(10):39–46.

Bastos, J. P., Geloso, V., and Bologna Pavlik, J. (2026). The forsaken road: Reassessing living standards following the cuban revolution and the american embargo. *Journal of Development Economics*. Forthcoming.

Bauer, T. K., Braun, S., and Kvasnicka, M. (2013). The economic integration of forced migrants: Evidence for post-war Germany. *Economic Journal*, 123(571):998–1024.

Becker, S. O. (2022). Forced displacement in history: Some recent research. *Australian Economic History Review*, 62(1):2–25.

Becker, S. O. and Ferrara, A. (2019). Consequences of forced migration: A survey of recent findings. *Labour Economics*, 59:1–16.

Becker, S. O., Grosfeld, I., Grosjean, P., Voigtländer, N., and Zhuravskaya, E. (2020). Forced migration and human capital: Evidence from post-WWII population transfers. *American Economic Review*, 110(5):1430–1463.

Becker, S. O., Lindenthal, V., Mukand, S., and Waldinger, F. (2024). Persecution and escape: Professional networks and high-skilled emigration from Nazi Germany. *American Economic Journal: Applied Economics*, 16(3):1–43.

Beerli, A., Ruffner, J., Siegenthaler, M., and Peri, G. (2021). The abolition of immigration restrictions and the performance of firms and workers: Evidence from Switzerland. *American Economic Review*, 111(3):976–1012.

Ben-Michael, E., Feller, A., and Rothstein, J. (2021). The augmented synthetic control method. *Journal of the American Statistical Association*, 116(536):1789–1803.

Bender, L. D. (1973). The Cuban exiles: An analytical sketch. *Journal of Latin American Studies*, 5(2):271–278.

Blum, M. and Rei, C. (2018). Escaping Europe: Health and human capital of Holocaust refugees. *European Review of Economic History*, 22(1):1–27.

Bodvarsson, O. B., Lewer, J. J., and Van den Berg, H. (2008). Measuring immigration's effects on labor demand: A reexamination of the Mariel boatlift. *Labour Economics*, 15(4):560–574.

Bologna Pavlik, J. and Young, A. T. (2026). Public participation and constitutional compliance. *American Journal of Political Science*, 70(2):824–841.

Borjas, G. J. (2017). The wage impact of the Marielitos: A reappraisal. *ILR Review*, 70(5):1077–1110.

Braun, S. and Mahmoud, T. O. (2014). The employment effects of immigration: Evidence from the mass arrival of German expellees in postwar Germany. *Journal of Economic History*, 74(1):69–108.

Braun, S. T. and Dwenger, N. (2020). Settlement location shapes the integration of forced migrants: Evidence from post-war Germany. *Explorations in Economic History*, 77:101330.

Buggle, J., Mayer, T., Sakalli, S. O., and Thoenig, M. (2023). The refugee's dilemma: Evidence from Jewish migration out of Nazi Germany. *Quarterly Journal of Economics*, 138(2):1273–1345.

Card, D. (1990). The impact of the Mariel boatlift on the Miami labor market. *Industrial and Labor Relations Review*, 43(2):245–257.

Castro Ruz, F. (1965). Discurso pronunciado en la clausura de la tercera plenaria nacional de la federación de mujeres cubanas, efectuada en isla de pinos. Portal Cuba. Speech delivered in Isla de Pinos, Cuba.

Cavallo, E., Galiani, S., Noy, I., and Pantano, J. (2013). Catastrophic Natural Disasters and Economic Growth. *The Review of Economics and Statistics*, 95(5):1549–1561.

Clarke, D., Pailañir, D., Athey, S., and Imbens, G. (2023). Synthetic difference in differences estimation. *IZA Discussion Paper No. 15907*. Stata implementation: `sdid`.

Clemens, M. A. and Hunt, J. (2019). The labor market effects of refugee waves: Reconciling conflicting results. *ILR Review*, 72(4):818–857.

Cobas, J. A., Aiken, M., and Jardine, D. F. (1993). Industrial segmentation, the ethnic economy, and job mobility: The case of Cuban exiles in Florida. *Quality & Quantity*, 27:249–270.

Devereux, J. (2021). The Absolution of History: Cuban Living Standards After 60 Years of Revolutionary Rule. *Revista de Historia Económica – Journal of Iberian and Latin American Economic History*, 39(1):5–36.

Dippel, C. and Heblich, S. (2021). Leadership in social movements: Evidence from the "forty-eighters" in the civil war. *American Economic Review*, 111(2):472–505.

Doeringer, P. B. and Piore, M. J. (1971). *Internal Labor Markets and Manpower Analysis*. D.C. Heath, Lexington, MA.

Domínguez, J. I. (1978). *Cuba: Order and Revolution*. Harvard University Press, Cambridge, MA.

Fagen, R. R. and Brody, R. A. (1964). Cubans in exile: A demographic analysis. *Social Problems*, 11(4):389–401.

Ferman, B., Pinto, C., and Possebom, V. (2020). Cherry picking with synthetic controls. *Journal of Policy Analysis and Management*, 39(2):510–532.

Florida Department of State, Division of Corporations (2026). Corporate data files. Bulk data download, Sunbiz.

Forment, C. A. (1989). Political practice and the rise of an ethnic enclave: The Cuban American case, 1959–1979. *Theory and Society*, 18(1):47–81.

Friedberg, R. M. (2001). The impact of mass migration on the Israeli labor market. *Quarterly Journal of Economics*, 116(4):1373–1408.

García, M. C. (1996). *Havana USA: Cuban Exiles and Cuban Americans in South Florida, 1959–1994*. University of California Press, Berkeley, CA.

Geloso, V. and Bologna Pavlik, J. (2021). The Cuban Revolution and Infant Mortality: A Synthetic Control Approach. *Explorations in Economic History*, 80:101376.

González Acosta, A. (2022). Apuntes para una historia de los estamentos aristocráticos cubanos: promotores y mecenas. *Estudios Nobiliarios y Emblemáticos de la Real Academia Matritense de Heráldica y Genealogía*, 2:57–85.

Grier, K. and Grier, R. (2021). The Washington consensus works: Causal effects of reform, 1970–2015. *Journal of Comparative Economics*, 49(1):59–72.

Guerra, L. (2018). *Heroes, Martyrs, and Political Messiahs in Revolutionary Cuba, 1946–1958*. Yale University Press, New Haven.

Hessami, Z., Schirner, S., and Wobbe, C. (2025). Asylum seekers, new businesses, and job creation. *Journal of Population Economics*, 38:87.

Hornung, E. (2014). Immigration and the diffusion of technology: The huguenot diaspora in prussia. *American Economic Review*, 104(1):84–122.

Hunt, J. (1992). The impact of the 1962 repatriates from Algeria on the French labor market. *Industrial and Labor Relations Review*, 45(3):556–572.

Hunt, J. and Gauthier-Loiselle, M. (2010). How much does immigration boost innovation? *American Economic Journal: Macroeconomics*, 2(2):31–56.

International Bank for Reconstruction and Development (1951). Report on Cuba: Findings and recommendations of an economic and technical mission. Technical report, International Bank for Reconstruction and Development, Washington, DC. Francis Adams Truslow, Chief of Mission.

Jahn, V. and Steinhardt, M. F. (2025). Immigration and new firm formation: Evidence from a quasi-experimental setting in Germany. *International Migration Review*, 59(2):963–995.

Jales, H., Kang, T. H., Stein, G., and Garcia Ribeiro, F. (2018). Measuring the role of the 1959 revolution on cuba's economic performance. *The World Economy*, 41(9):2243–2274.

Jiménez Soler, G. (2008). *Los propietarios de Cuba 1958*. Editorial de Ciencias Sociales, La Habana.

Kerr, S. P. and Kerr, W. R. (2020). Immigrant entrepreneurship in America: Evidence from the survey of business owners 2007 & 2012. *Research Policy*, 49(3):103918.

Kerr, W. R. and Lincoln, W. F. (2010). The supply side of innovation: H-1B visa reforms and U.S. ethnic invention. *Journal of Labor Economics*, 28(3):473–508.

Kirzner, I. M. (1973). *Competition and Entrepreneurship*. University of Chicago Press, Chicago.

Lee, S. and Lin, J. (2018). Natural amenities, neighbourhood dynamics, and persistence in the spatial distribution of income. *Review of Economic Studies*, 85(1):663–694.

Leuven, E. and Sianesi, B. (2003). PSMATCH2: Stata module to perform full Mahalanobis and propensity score matching, common support graphing, and covariate imbalance testing. Statistical Software Components S432001, Boston College Department of Economics.

Library of Congress (2025). U.s. telephone directory collection. Digitized telephone directories, Library of Congress.

Locay, L. and Sanguinetty, J. (1996). The effect of socialism on the entrepreneurial abilities of Cuban-Americans. In *Cuba in Transition — Volume 6*, pages 320–327. Association for the Study of the Cuban Economy, Washington, DC.

Manson, S., Schroeder, J., Van Riper, D., Kugler, T., and Ruggles, S. (2026). IPUMS National Historical Geographic Information System: Version 21.0 [dataset]. Minneapolis, MN: IPUMS.

Martinez, C. and Geloso, V. (2026). Dirty commies: The effect of the Cuban revolution in greenhouse gas emissions. SSRN Working Paper 7043339, Social Science Research Network.

Mesa-Lago, C. (1981). *The Economy of Socialist Cuba: A Two-Decade Appraisal*. University of New Mexico Press, Albuquerque, NM.

Mitchell, W. L. (1962). The Cuban refugee program. *Social Security Bulletin*, 25(3):3–8.

Moser, P., Voena, A., and Waldinger, F. (2014). German jewish émigrés and US invention. *American Economic Review*, 104(10):3222–3255.

Nakamura, E., Sigurdsson, J., and Steinsson, J. (2022). The gift of moving: Intergenerational consequences of a mobility shock. *Review of Economic Studies*, 89(3):1557–1592.

Ochsner, C. and Roesel, F. (2020). Migrating extremists. *Economic Journal*, 130(628):1135–1172.

Pedraza-Bailey, S. (1985). Cuba's exiles: Portrait of a refugee migration. *International Migration Review*, 19(1):4–34.

Pérez, L. (1986). Cubans in the United States. *The ANNALS of the American Academy of Political and Social Science*, 487:126–137.

Peri, G. and Yasenov, V. (2019). The labor market effects of a refugee wave: Synthetic control method meets the Mariel boatlift. *Journal of Human Resources*, 54(2):267–309.

Peters, M. (2022). Market size and spatial growth—evidence from Germany's post-war population expulsions. *Econometrica*, 90(5):2357–2396.

Portes, A. (1969). Dilemmas of a golden exile: Integration of Cuban refugee families in Milwaukee. *American Sociological Review*, 34(4):505–518.

Portes, A. (1987). The social origins of the Cuban enclave economy of Miami. *Sociological Perspectives*, 30(4):340–372.

Portes, A. and Puhrmann, A. (2015). A bifurcated enclave: The economic evolution of the Cuban and Cuban American population of metropolitan Miami. *Cuban Studies*, 43:40–63.

Portes, A. and Stepick, A. (1985). Unwelcome immigrants: The labor market experiences of 1980 (Mariel) Cuban and Haitian refugees in South Florida. *American Sociological Review*, 50(4):493–514.

Portes, A. and Zhou, M. (1996). Self-employment and the earnings of immigrants. *American Sociological Review*, 61(2):219–230.

Pérez, L. (2023). *The House on G Street: A Cuban Family Saga*. New York University Press, New York.

Roorda, E. P. (2004). Desarraigando la tierra de clubes: la extinción de la «colonia americana» en la habana. *Secuencia*, (60):111–134.

Sarvimäki, M., Uusitalo, R., and Jäntti, M. (2022). Habit formation and the misallocation of labor: Evidence from forced migrations. *Journal of the European Economic Association*, 20(6):2497–2539.

Thomas, H. (1977). *The Cuban Revolution*. Harper & Row, New York. Abridged edition of *Cuba: The Pursuit of Freedom* (1971).

Thomas, J. F. (1967). Cuban refugees in the United States. *International Migration Review*, 1(2):46–57.

Toews, G. and Vézina, P.-L. (2025). Enemies of the people. *American Economic Journal: Macroeconomics*, 17(1):310–342.

United Nations, Population Division, D. o. E. a. S. A. (2024). World Population Prospects 2024.

Verme, P. and Schuettler, K. (2021). The impact of forced displacement on host communities: A review of the empirical literature in economics. *Journal of Development Economics*, 150:102606.

Vázquez, A. and Pombo, O. (2013). ¿alta costura en cuba?: bosquejo sociocultural de un silencio. *Islas*, 55(174):44–55.

Vázquez Fleitas, L. (2023). Del «matrimonio» y otros «ritos de vida burgueses»: los rituales de afirmación de la élite socioeconómica cubana y su representación en la crónica social (1947–1958). *Revista de Historia Social y de las Mentalidades*, 27(2):159–207.

Vázquez Fleitas, L. (2024). Había una vez la habana: moderna, burguesa, cosmopolita y excluyente. la élite burguesa habanera y la urbanización de sus espacios sociales (1940–1958). *Revista de la Red de Intercátedras de Historia de América Latina Contemporánea*, 20:66–101.

Ward, M. and Devereux, J. (2012). The road not taken: Pre-revolutionary Cuban living standards in comparative perspective. *The Journal of Economic History*, 72(1):104–132.

Wenk, M. G. (1968). Adjustment and assimilation: The Cuban refugee experience. *International Migration Review*, 3(1):38–49.

Wilson, K. L. and Martin, W. A. (1982). Ethnic enclaves: A comparison of the Cuban and black economies in Miami. *American Journal of Sociology*, 88(1):135–160.

Wilson, K. L. and Portes, A. (1980). Immigrant enclaves: An analysis of the labor market experiences of Cubans in Miami. *American Journal of Sociology*, 86(2):295–319.

Wiltshire, J. C. (2022). ALLSYNTH: Stata module to automate estimation of (i) bias-corrected synthetic control gaps. Statistical Software Components S459076, Boston College Department of Economics. Revised 10 Nov 2024; forthcoming, *The Stata Journal*.

Yao, L., Bolen, J. B., and Williamson, C. R. (2022). Are economic arguments against immigration missing the boat? the fiscal effects of the Mariel Boatlift. *Southern Economic Journal*, 89(2):305–325.

Álvarez de Cañas, P., de Posada, J., and Cidre, E., editors (1958). *Libro de Oro de la Sociedad Habanera*. Talleres Tipográficos de la Editorial Lex, La Habana. Annual social register of Havana. Editions consulted: 1947–1960.

Tables

Table 1: Occupations of the Havana elite and of the exiles matched to Florida, 1958

	Havana elite		**Exiles**		
Occupation of household head	*n*	%	*n*	%	Diff.
Business owner or executive	547	16.3	100	18.8	+2.5
Agriculture and sugar	91	2.7	19	3.6	+0.9
Law	753	22.5	112	21.1	−1.4
Medicine	651	19.4	97	18.3	−1.2
Engineering and architecture	304	9.1	49	9.2	+0.2
Accounting and finance	111	3.3	28	5.3	+2.0
Other professional	71	2.1	10	1.9	−0.2
Politics and military	70	2.1	13	2.4	+0.4
Property only, no stated occupation	327	9.8	46	8.7	−1.1
Other / not classifiable	424	12.7	57	10.7	−1.9
Households with an occupation recorded	3,349	62.6	531	63.7	
Households in source	5,354		833		

Occupational composition of every household in the *Libro de Oro de la Sociedad Habanera* (1958), the register of the pre-revolutionary Havana elite, beside the subset whose members are matched to post-1959 Florida telephone directories by the record-linkage procedure. "Exiles" are the matched households. Percentages are of households with an occupation recorded (3,349 and 531); the final column is the exile share minus the Havana share, in percentage points.

Table 2: Summary statistics, census tracts

	Treated $N = 23$		Dade pool $N = 47$		Comparable metros $N = 2,234$	
	Mean	S.D.	Mean	S.D.	Mean	S.D.
Matching covariates, 1960						
Log median family income	8.531	0.250	8.589	0.315	8.799	0.316
Median school years	11.521	0.932	10.932	1.697	11.172	1.653
Spanish-surname share	0.014	0.013	0.006	0.007	0.022	0.070
Professional and managerial share	0.259	0.098	0.226	0.140	0.232	0.133
Log population density	8.024	0.381	7.420	1.104	7.118	1.366
Outcome levels, 1950						
Unemployment rate	0.059	0.018	0.050	0.017	0.038	0.025
Employment/population	0.414	0.040	0.372	0.055	0.391	0.056
Log median family income	7.996	0.219	7.969	0.248	8.105	0.293
Outcome levels, 1960						
Unemployment rate	0.061	0.023	0.052	0.027	0.042	0.030
Employment/population	0.433	0.032	0.385	0.036	0.387	0.065
Log median family income	8.531	0.250	8.589	0.315	8.799	0.316
Outcome changes, 1960–70						
Unemployment rate	-0.021	0.019	-0.018	0.022	0.000	0.025
Employment/population	0.026	0.049	0.032	0.067	0.020	0.051
Log median family income	0.477	0.080	0.509	0.142	0.456	0.162
Outcome changes, 1960–80						
Unemployment rate	-0.009	0.014	0.002	0.022	0.024	0.041
Employment/population	0.040	0.077	0.009	0.167	0.059	0.102
Log median family income	1.108	0.144	1.107	0.237	1.094	0.240

Treated tracts are the 23 Dade County tracts receiving at least five elite households. Control pools are untreated tracts in the rest of Dade County and in the 26 comparable metropolitan counties, those with at least 20 tracts and 50,000 people in 1950 and similar population and population growth to Miami. Tracts receiving one to four elite households are excluded from both pools. Values from 1950, 1970 and 1980 are harmonized to 1960 tract boundaries.

Table 3: Effect of elite settlement on census-tract outcomes

Outcome	Dade County (47 controls)	Comparable metros (2,234 controls)
Unemployment rate		
Δ1960–70	+0.006 (4/9)	−0.018* (5/9)
	[+0.001, +0.013]	[−0.021, −0.002]
Δ1960–80	−0.023* (5/9)	−0.019* (7/9)
	[−0.030, −0.016]	[−0.031, −0.013]
Δ1950–60 *(placebo)*	+0.005 (0/9)	+0.005 (0/9)
	[+0.001, +0.011]	[+0.001, +0.009]
Employment/population		
Δ1960–70	+0.034 (2/9)	+0.017 (3/9)
	[+0.013, +0.047]	[+0.006, +0.037]
Δ1960–80	+0.048 (3/9)	+0.004 (2/9)
	[−0.034, +0.084]	[−0.016, +0.051]
Δ1950–60 *(placebo)*	+0.021 (3/9)	+0.015* (6/9)
	[−0.001, +0.082]	[+0.012, +0.023]
Log median family income		
Δ1960–70	−0.040 (1/9)	+0.043 (4/9)
	[−0.053, −0.012]	[+0.018, +0.063]
Δ1960–80	−0.036 (1/9)	+0.090* (5/9)
	[−0.137, +0.217]	[+0.025, +0.118]
Δ1950–60 *(placebo)*	+0.017 (0/9)	−0.205* (9/9)
	[−0.077, +0.060]	[−0.254, −0.144]

Median average treatment effect on the treated across the nine matching estimators, with the number significant at 10% in parentheses and the minimum–maximum across estimators in brackets. * marks a majority of the nine. Estimators: `psmatch2` propensity-score matching at one to four nearest neighbours plus a normal kernel, with common support and a 250-replication bootstrap, and `teffects nnmatch` Mahalanobis matching at one to four neighbours with Abadie–Imbens bias adjustment. The placebo replaces the outcome with the 1950–60 change and matches on 1950 characteristics only.

Figures

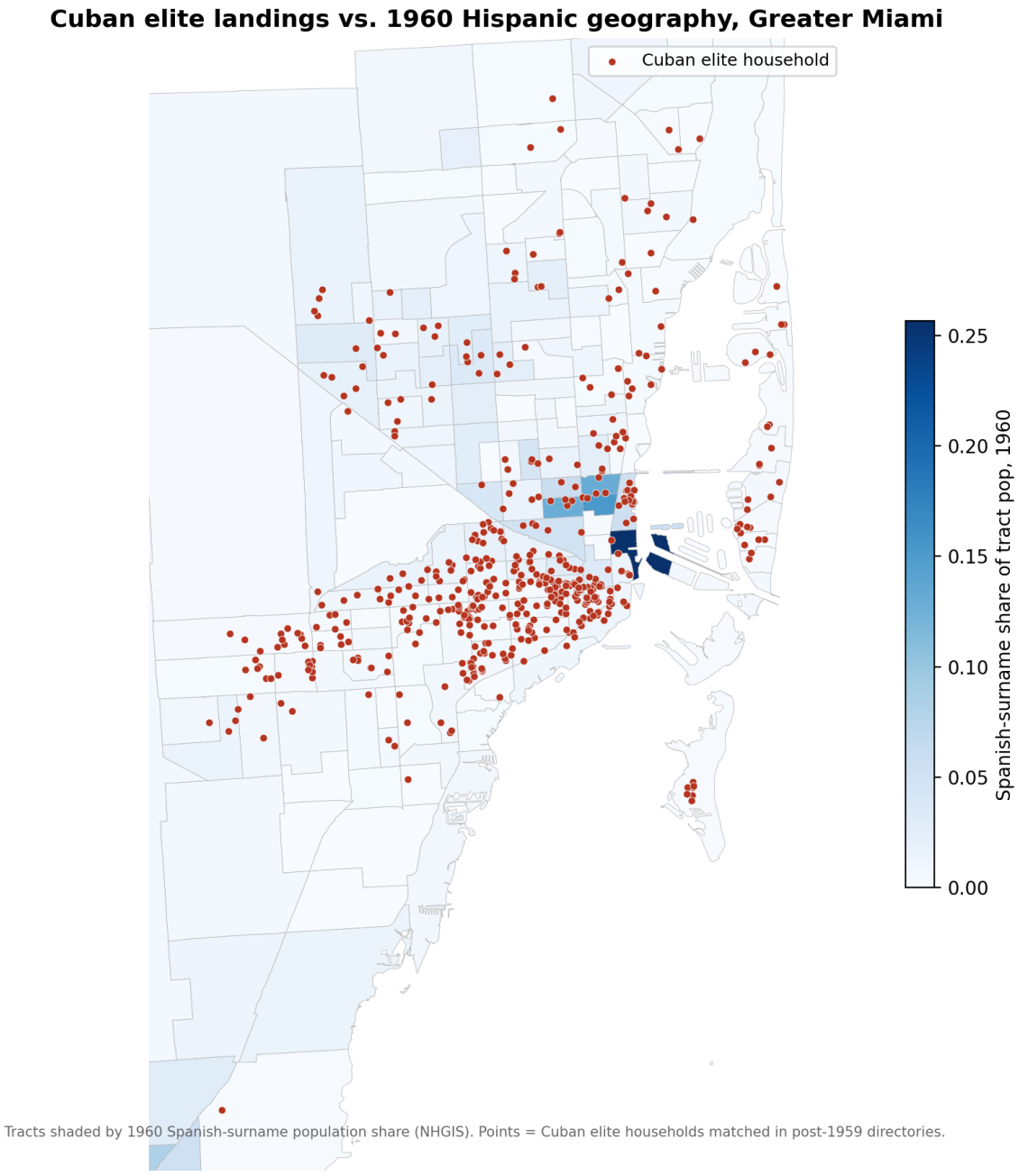


Figure 1: Elite settlement and the pre-existing Hispanic geography of Miami. Elite households located from the 1959–64 phone directories, aggregated to 1960 census tracts, against the 1960 Spanish-surname population share.

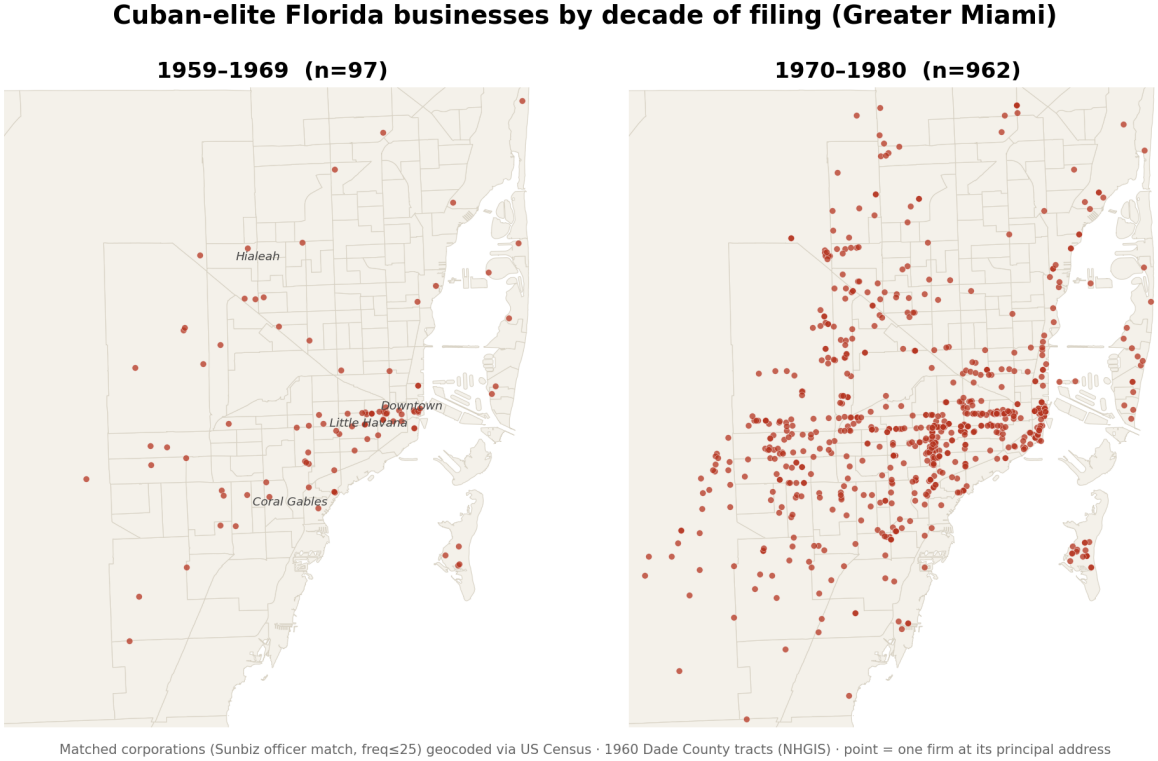


Figure 2: Businesses owned by the exiled elite, by decade of first appearance. Firms are matched from the Florida corporate registry and the city directories to the linked elite roster.

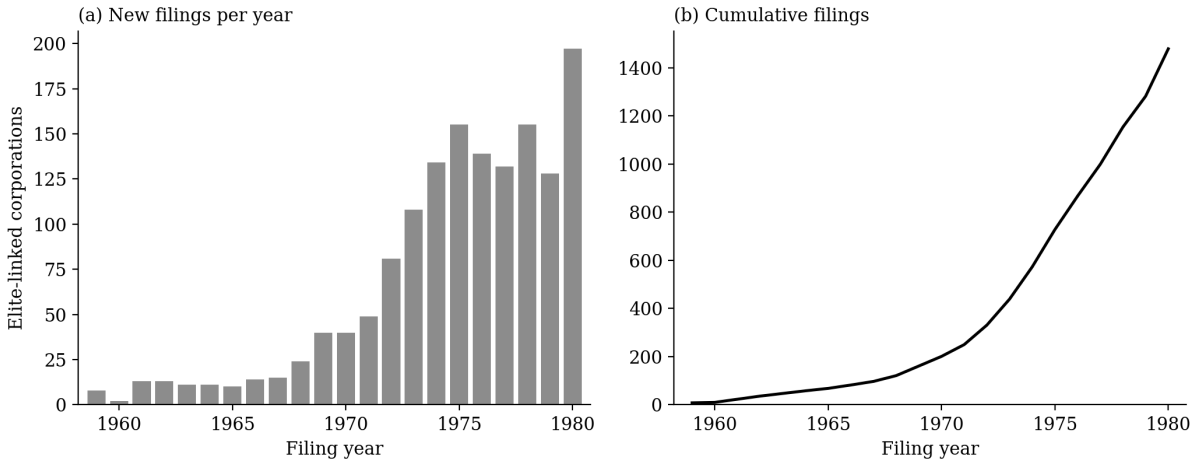


Figure 3: Elite-linked corporations by filing year, 1959–80. Panel (a) shows new filings per year and panel (b) the cumulative total, for the 1,479 Florida corporations filed from 1959 onward whose officer records match the linked elite roster.

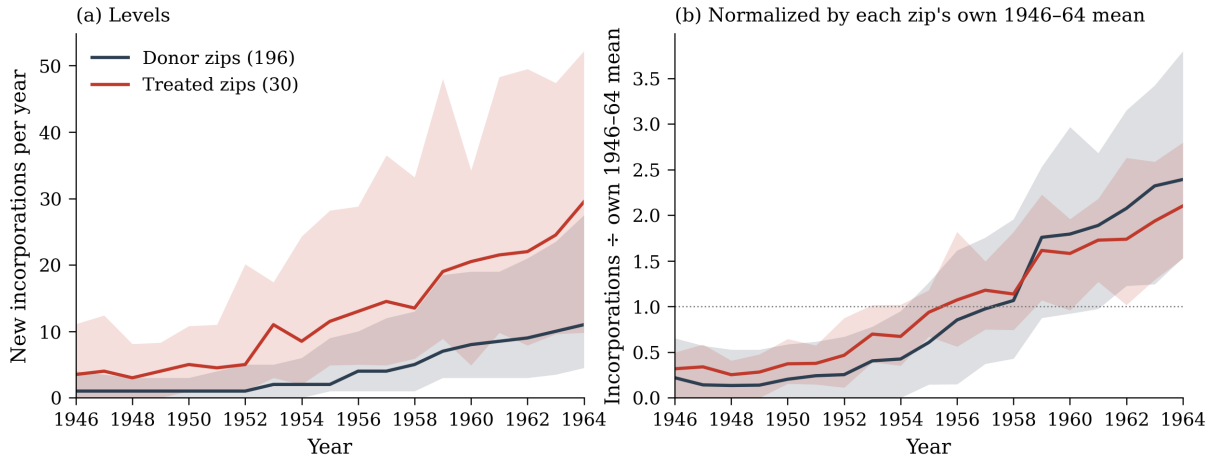


Figure 4: Treated and donor zip codes before treatment, 1946–64. (a) New incorporations per zip code per year. (b) Each zip code's incorporations divided by its own 1946–64 mean. Lines are medians across zip codes and shaded bands span the 10th to 90th percentiles.

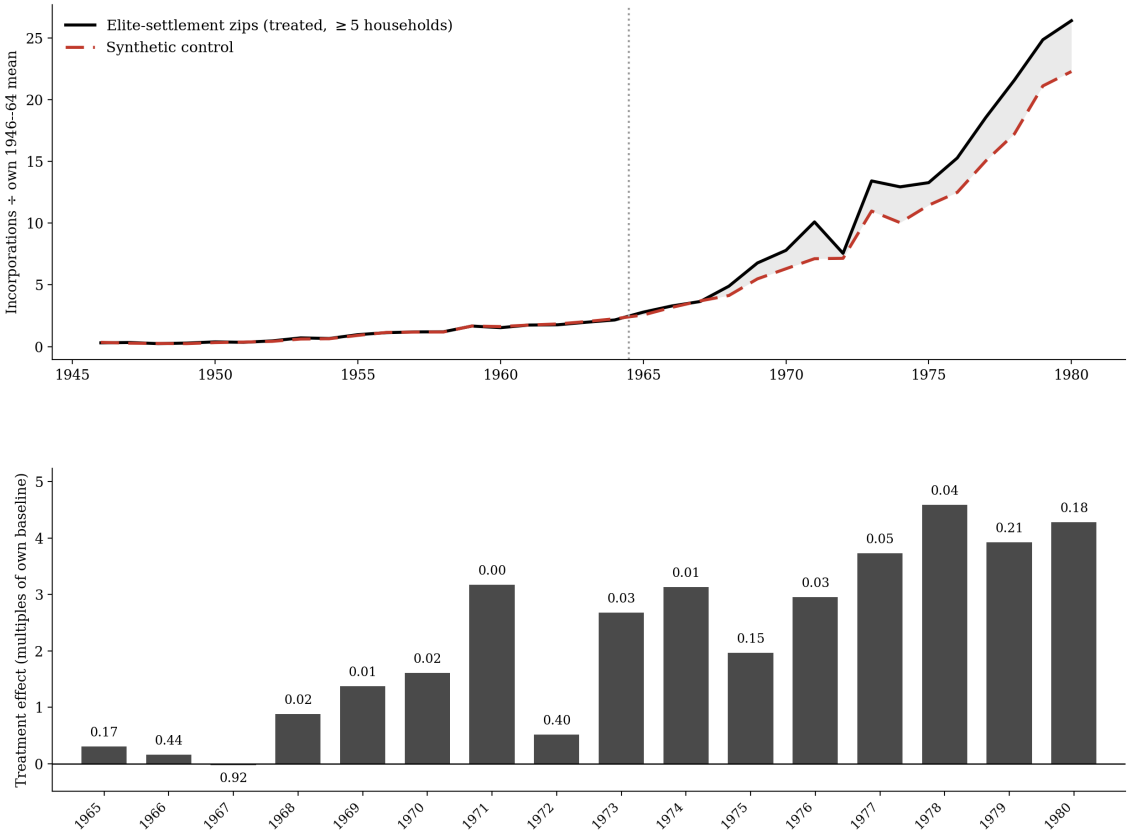


Figure 5: Synthetic control estimates of business formation with treatment dated 1965. The outcome is each zip code's new Florida incorporations divided by its own 1946–58 mean. Top panel: mean treated and synthetic series. Bottom panel: effect by year, with standardized permutation-based p-values from 100,000 placebo bundles.

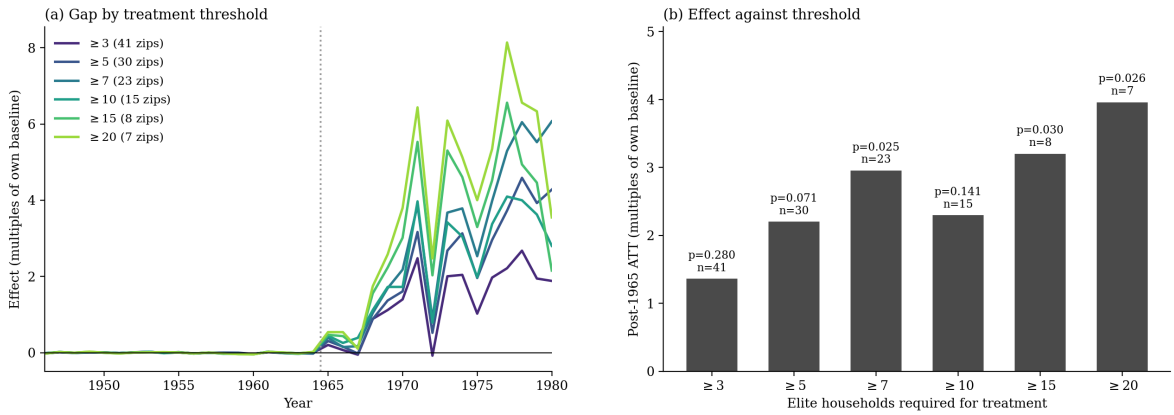


Figure 6: Threshold variation. Every threshold reuses the same per-unit synthetic controls, fitted once against the same 196 donor zip codes, so raising it only changes which treated zips enter the average.

10 Appendix

The matching results, outcome by outcome

Each table reports all nine matching estimators for one outcome against one control pool, at both treatment horizons and at the placebo, together with the covariate balance χ^2 after matching.

Table A1: Unemployment rate, Dade County: estimates and covariate balance

	Δ1960–70		Δ1960–80		Δ1950–60 *(placebo)*	
Estimator	ATT	χ^2	ATT	χ^2	ATT	χ^2
Propensity score, 1 NN	+0.006 (0.016)	15.0 [0.04]	−0.030 (0.021)	10.2 [0.18]	+0.002 (0.015)	—
Propensity score, 2 NN	+0.001 (0.012)	4.0 [0.78]	−0.018 (0.017)	1.9 [0.97]	+0.001 (0.013)	2.0 [0.85]
Propensity score, 3 NN	+0.004 (0.010)	2.6 [0.92]	−0.017 (0.015)	1.6 [0.98]	+0.005 (0.012)	0.8 [0.98]
Propensity score, 4 NN	+0.004 (0.009)	2.7 [0.91]	−0.016 (0.014)	2.5 [0.93]	+0.002 (0.011)	0.3 [1.00]
Propensity score, kernel	+0.008 (0.009)	3.3 [0.86]	−0.027* (0.016)	3.1 [0.88]	+0.004 (0.010)	0.6 [0.99]
Mahalanobis, 1 NN	+0.013*** (0.004)	n.a.	−0.019** (0.008)	n.a.	+0.011 (0.009)	n.a.
Mahalanobis, 2 NN	+0.009*** (0.003)	n.a.	−0.023*** (0.007)	n.a.	+0.011 (0.009)	n.a.
Mahalanobis, 3 NN	+0.007** (0.003)	n.a.	−0.027*** (0.006)	n.a.	+0.009 (0.009)	n.a.
Mahalanobis, 4 NN	+0.006* (0.003)	n.a.	−0.028*** (0.006)	n.a.	+0.009 (0.009)	n.a.

Average treatment effect on the treated, standard error in parentheses; ***, ** and * mark significance at 1%, 5% and 10%. The χ^2 column tests the null that the matching covariates are balanced between treated tracts and their matched controls after matching, with its p-value in brackets, reported for the propensity-score estimators only, since `pstest` requires `psmatch2`'s weights, which `teffects nnmatch` does not create.

Table A2: Unemployment rate, comparable metros: estimates and covariate balance

	Δ1960–70		Δ1960–80		Δ1950–60 *(placebo)*	
Estimator	ATT	χ^2	ATT	χ^2	ATT	χ^2
Propensity score, 1 NN	−0.002 (0.015)	6.9 [0.44]	−0.017 (0.015)	5.7 [0.57]	+0.008 (0.009)	2.1 [0.83]
Propensity score, 2 NN	−0.007 (0.012)	4.6 [0.71]	−0.023* (0.012)	4.7 [0.69]	+0.009 (0.007)	2.4 [0.80]
Propensity score, 3 NN	−0.010 (0.011)	3.4 [0.85]	−0.023** (0.011)	4.1 [0.76]	+0.007 (0.006)	4.4 [0.49]
Propensity score, 4 NN	−0.012 (0.011)	3.4 [0.85]	−0.027*** (0.010)	3.2 [0.87]	+0.005 (0.006)	4.8 [0.44]
Propensity score, kernel	−0.021*** (0.004)	20.5 [0.00]	−0.031*** (0.004)	20.5 [0.00]	+0.001 (0.003)	19.3 [0.00]
Mahalanobis, 1 NN	−0.018*** (0.006)	n.a.	−0.015* (0.008)	n.a.	+0.006 (0.004)	n.a.
Mahalanobis, 2 NN	−0.021*** (0.006)	n.a.	−0.013 (0.009)	n.a.	+0.003 (0.006)	n.a.
Mahalanobis, 3 NN	−0.021*** (0.005)	n.a.	−0.015** (0.007)	n.a.	+0.001 (0.004)	n.a.
Mahalanobis, 4 NN	−0.018*** (0.004)	n.a.	−0.019*** (0.006)	n.a.	+0.001 (0.005)	n.a.

Average treatment effect on the treated, standard error in parentheses; ***, ** and * mark significance at 1%, 5% and 10%. The χ^2 column tests the null that the matching covariates are balanced between treated tracts and their matched controls after matching, with its p-value in brackets, reported for the propensity-score estimators only, since `pstest` requires `psmatch2`'s weights, which `teffects nnmatch` does not create.

Table A3: Employment/population, Dade County: estimates and covariate balance

	Δ1960–70		Δ1960–80		Δ1950–60 *(placebo)*	
Estimator	ATT	χ^2	ATT	χ^2	ATT	χ^2
Propensity score, 1 NN	+0.013 (0.047)	2.1 [0.96]	−0.034 (0.096)	2.1 [0.96]	+0.012 (0.089)	5.7 [0.33]
Propensity score, 2 NN	+0.031 (0.043)	2.5 [0.92]	+0.010 (0.084)	2.5 [0.92]	+0.065 (0.082)	3.7 [0.60]
Propensity score, 3 NN	+0.047 (0.036)	4.1 [0.77]	+0.048 (0.075)	4.1 [0.77]	+0.082 (0.066)	3.3 [0.65]
Propensity score, 4 NN	+0.045 (0.033)	5.0 [0.66]	+0.047 (0.068)	5.0 [0.66]	+0.069 (0.063)	1.7 [0.89]
Propensity score, kernel	+0.034 (0.030)	—	+0.017 (0.073)	—	+0.022 (0.019)	0.9 [0.97]
Mahalanobis, 1 NN	+0.047*** (0.018)	n.a.	+0.067 (0.045)	n.a.	−0.001 (0.014)	n.a.
Mahalanobis, 2 NN	+0.041** (0.019)	n.a.	+0.073** (0.037)	n.a.	+0.017** (0.009)	n.a.
Mahalanobis, 3 NN	+0.015 (0.013)	n.a.	+0.084** (0.036)	n.a.	+0.021** (0.008)	n.a.
Mahalanobis, 4 NN	+0.016 (0.013)	n.a.	+0.071* (0.037)	n.a.	+0.015* (0.009)	n.a.

Average treatment effect on the treated, standard error in parentheses; ***, ** and * mark significance at 1%, 5% and 10%. The χ^2 column tests the null that the matching covariates are balanced between treated tracts and their matched controls after matching, with its p-value in brackets, reported for the propensity-score estimators only, since `pstest` requires `psmatch2`'s weights, which `teffects nnmatch` does not create.

Table A4: Employment/population, comparable metros: estimates and covariate balance

	Δ1960–70		Δ1960–80		Δ1950–60 *(placebo)*	
Estimator	ATT	χ^2	ATT	χ^2	ATT	χ^2
Propensity score, 1 NN	+0.025 (0.021)	34.1 [0.00]	+0.042 (0.032)	28.5 [0.00]	+0.016 (0.015)	47.2 [0.00]
Propensity score, 2 NN	+0.032* (0.018)	15.9 [0.03]	+0.045 (0.027)	15.9 [0.03]	+0.015 (0.013)	43.9 [0.00]
Propensity score, 3 NN	+0.037** (0.017)	11.0 [0.14]	+0.045* (0.025)	10.9 [0.14]	+0.015 (0.011)	24.5 [0.00]
Propensity score, 4 NN	+0.037** (0.016)	11.3 [0.13]	+0.051** (0.023)	11.2 [0.13]	+0.017* (0.010)	26.2 [0.00]
Propensity score, kernel	+0.007 (0.010)	33.7 [0.00]	−0.016 (0.016)	33.7 [0.00]	+0.023*** (0.006)	29.2 [0.00]
Mahalanobis, 1 NN	+0.006 (0.016)	n.a.	+0.004 (0.018)	n.a.	+0.019*** (0.006)	n.a.
Mahalanobis, 2 NN	+0.011 (0.015)	n.a.	−0.002 (0.016)	n.a.	+0.013*** (0.004)	n.a.
Mahalanobis, 3 NN	+0.017 (0.012)	n.a.	−0.010 (0.015)	n.a.	+0.012*** (0.005)	n.a.
Mahalanobis, 4 NN	+0.015 (0.011)	n.a.	−0.003 (0.013)	n.a.	+0.012** (0.005)	n.a.

Average treatment effect on the treated, standard error in parentheses; ***, ** and * mark significance at 1%, 5% and 10%. The χ^2 column tests the null that the matching covariates are balanced between treated tracts and their matched controls after matching, with its p-value in brackets, reported for the propensity-score estimators only, since `pstest` requires `psmatch2`'s weights, which `teffects nnmatch` does not create.

Table A5: Log median family income, Dade County: estimates and covariate balance

	Δ1960–70		Δ1960–80		Δ1950–60 *(placebo)*	
Estimator	ATT	χ^2	ATT	χ^2	ATT	χ^2
Propensity score, 1 NN	−0.040 (0.106)	1.5 [0.96]	−0.053 (0.231)	1.5 [0.96]	−0.069 (0.145)	2.4 [0.66]
Propensity score, 2 NN	−0.012 (0.097)	1.4 [0.96]	−0.034 (0.181)	0.9 [0.99]	+0.038 (0.119)	0.8 [0.94]
Propensity score, 3 NN	−0.042 (0.087)	1.2 [0.97]	−0.108 (0.163)	0.9 [0.99]	+0.060 (0.110)	1.0 [0.90]
Propensity score, 4 NN	−0.051 (0.082)	1.1 [0.98]	−0.137 (0.158)	0.6 [1.00]	+0.045 (0.102)	1.5 [0.83]
Propensity score, kernel	−0.053 (0.084)	0.3 [1.00]	−0.078 (0.181)	0.3 [1.00]	+0.029 (0.093)	1.1 [0.89]
Mahalanobis, 1 NN	−0.018 (0.027)	n.a.	+0.217** (0.103)	n.a.	−0.031 (0.074)	n.a.
Mahalanobis, 2 NN	−0.047* (0.025)	n.a.	+0.053 (0.065)	n.a.	+0.017 (0.061)	n.a.
Mahalanobis, 3 NN	−0.015 (0.028)	n.a.	−0.036 (0.058)	n.a.	−0.021 (0.060)	n.a.
Mahalanobis, 4 NN	−0.018 (0.028)	n.a.	−0.001 (0.057)	n.a.	−0.077 (0.061)	n.a.

Average treatment effect on the treated, standard error in parentheses; ***, ** and * mark significance at 1%, 5% and 10%. The χ^2 column tests the null that the matching covariates are balanced between treated tracts and their matched controls after matching, with its p-value in brackets, reported for the propensity-score estimators only, since `pstest` requires `psmatch2`'s weights, which `teffects nnmatch` does not create.

Table A6: Log median family income, comparable metros: estimates and covariate balance

	Δ1960–70		Δ1960–80		Δ1950–60 *(placebo)*	
Estimator	ATT	χ^2	ATT	χ^2	ATT	χ^2
Propensity score, 1 NN	+0.032 (0.058)	7.2 [0.30]	+0.098 (0.073)	5.7 [0.45]	−0.205*** (0.076)	6.2 [0.18]
Propensity score, 2 NN	+0.018 (0.051)	3.1 [0.79]	+0.118* (0.069)	2.9 [0.82]	−0.169** (0.071)	10.6 [0.03]
Propensity score, 3 NN	+0.043 (0.045)	3.1 [0.79]	+0.110* (0.064)	2.0 [0.92]	−0.144** (0.065)	13.2 [0.01]
Propensity score, 4 NN	+0.044 (0.041)	1.7 [0.95]	+0.095 (0.060)	1.6 [0.95]	−0.159*** (0.058)	7.7 [0.10]
Propensity score, kernel	+0.027 (0.019)	29.8 [0.00]	+0.025 (0.037)	29.8 [0.00]	−0.151*** (0.028)	—
Mahalanobis, 1 NN	+0.063** (0.026)	n.a.	+0.083 (0.051)	n.a.	−0.220*** (0.040)	n.a.
Mahalanobis, 2 NN	+0.059** (0.023)	n.a.	+0.090** (0.045)	n.a.	−0.213*** (0.035)	n.a.
Mahalanobis, 3 NN	+0.055** (0.023)	n.a.	+0.065* (0.039)	n.a.	−0.237*** (0.034)	n.a.
Mahalanobis, 4 NN	+0.042* (0.024)	n.a.	+0.062* (0.037)	n.a.	−0.254*** (0.032)	n.a.

Average treatment effect on the treated, standard error in parentheses; ***, ** and * mark significance at 1%, 5% and 10%. The χ^2 column tests the null that the matching covariates are balanced between treated tracts and their matched controls after matching, with its p-value in brackets, reported for the propensity-score estimators only, since `pstest` requires `psmatch2`'s weights, which `teffects nnmatch` does not create.

Robustness checks

10.0.1 Synthetic Difference-in-differences

Table A7: Business formation: synthetic difference-in-differences

Estimator	ATT	s.e.	t	95% CI
Synthetic difference-in-differences	**+39.89**	**15.04**	**+2.65**	**[+10.4, +69.4]**

Outcome: new Florida incorporations per zip code per year, in levels, 1946–80, treatment 1965. 30 treated zip codes against the same 196 donor zip codes the synthetic control uses. The bootstrap uses 500 replications, seed 1234.

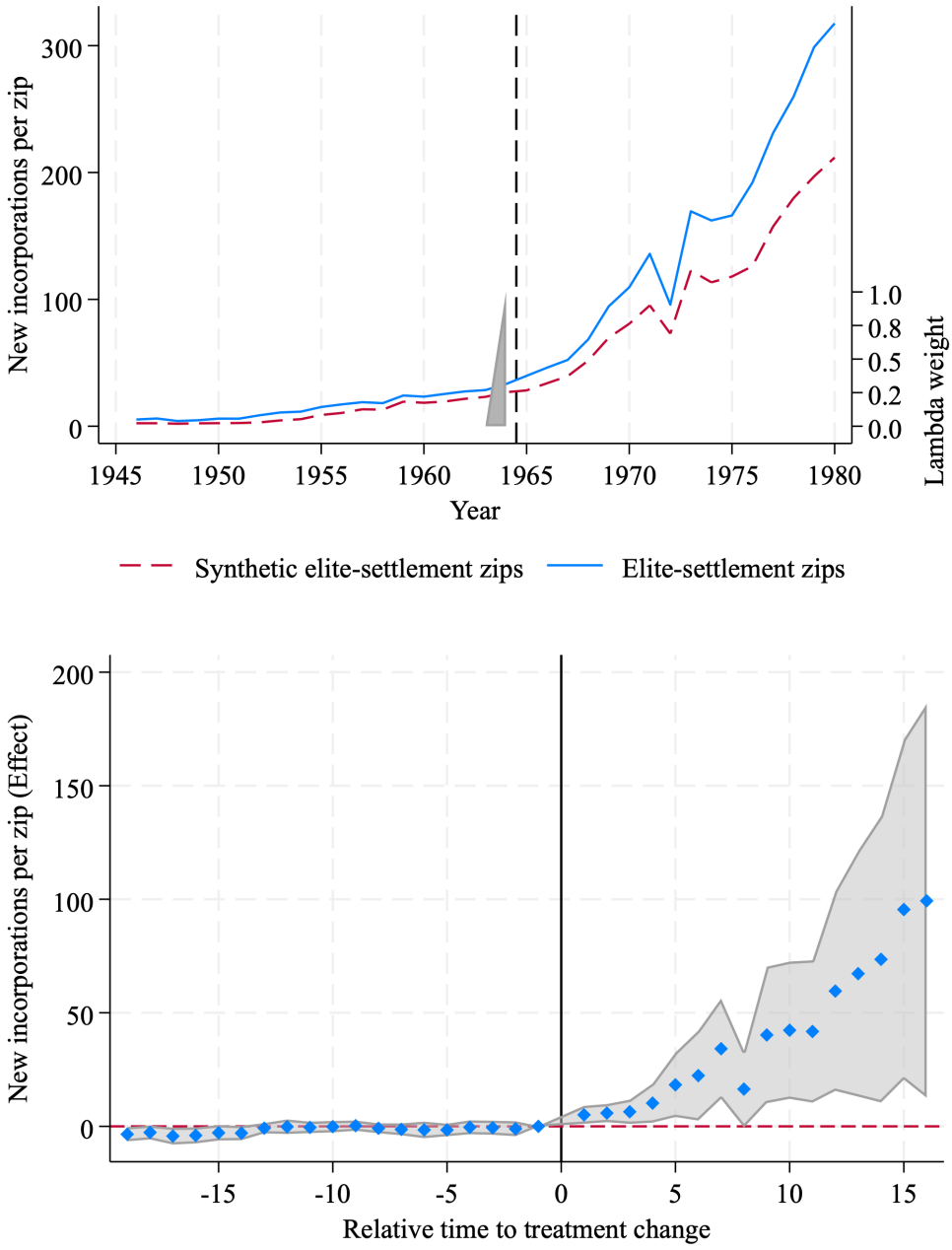


Figure A1: Synthetic difference-in-differences, new incorporations per zip code per year. Upper panel: treated and synthetic series, with the estimator's time weights λ shaded below. Lower panel: the `sdid_event` study in relative time, with 95% bootstrap confidence bands from 500 replications.

10.0.2 Business formation with the 1959 treatment date

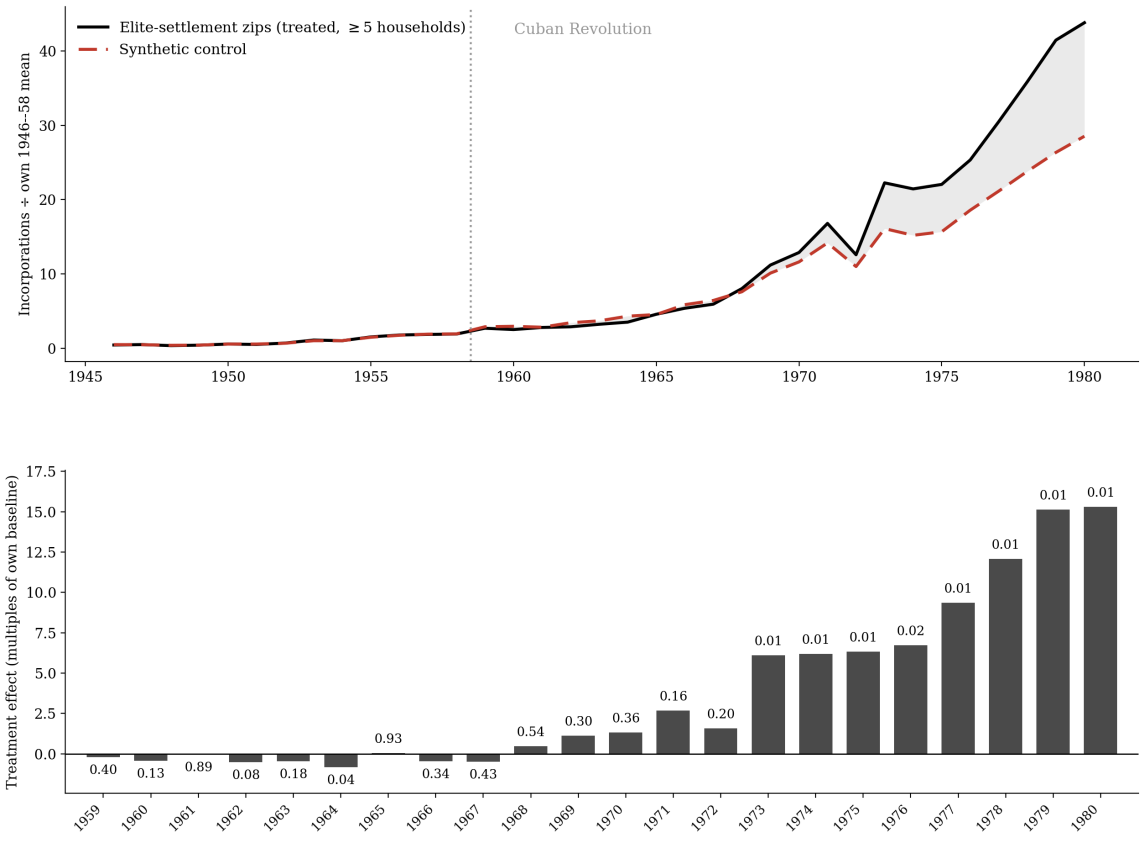


Figure A2: Synthetic control estimates of business formation with treatment dated 1959. The outcome is each zip code's new Florida incorporations divided by its own 1946–58 mean. Top panel: mean treated and synthetic series. Bottom panel: effect by year, with standardized permutation-based p-values from 100,000 placebo bundles.

10.0.3 Higher and lower treatment thresholds

Table A8: Robustness: the tract design at higher and lower treatment thresholds

Outcome	Threshold	Dade	Comparable metros
Panel A. Δ1960–70			
Unemployment rate	≥3	+0.006 (4/9)	-0.020* (9/9)
	≥5	+0.006 (4/9)	-0.018* (5/9)
	≥7	+0.009 (4/9)	-0.020* (5/9)
Employment/population	≥3	+0.005 (2/9)	+0.004 (4/9)
	≥5	+0.034 (2/9)	+0.017 (3/9)
	≥7	+0.014 (2/9)	+0.027* (5/9)
Median family income (log)	≥3	-0.036 (2/9)	+0.044* (6/9)
	≥5	-0.040 (1/9)	+0.043 (4/9)
	≥7	-0.043 (0/9)	+0.090* (7/9)
Panel B. Δ1960–80			
Unemployment rate	≥3	-0.020* (7/9)	-0.022* (9/9)
	≥5	-0.023* (5/9)	-0.019* (7/9)
	≥7	-0.024 (4/9)	-0.026* (8/9)
Employment/population	≥3	+0.012 (3/9)	-0.009 (3/9)
	≥5	+0.048 (3/9)	+0.004 (2/9)
	≥7	-0.007 (2/9)	+0.030 (3/9)
Median family income (log)	≥3	-0.036 (0/9)	+0.052 (4/9)
	≥5	-0.036 (1/9)	+0.090* (5/9)
	≥7	-0.023 (1/9)	+0.089* (5/9)

Median average treatment effect on the treated across the nine matching estimators, with the number significant at 10% in parentheses and the minimum–maximum across estimators in brackets. * marks a majority of the nine. Estimators: `psmatch2` propensity-score matching at one to four nearest neighbours plus a normal kernel, with common support and a 250-replication bootstrap, and `teffects nnmatch` Mahalanobis matching at one to four neighbours with Abadie–Imbens bias adjustment. The placebo replaces the outcome with the 1950–60 change and matches on 1950 characteristics only.

"Threshold" is the number of elite households a tract must receive to count as treated. The adopted specification uses ≥5, giving 23 treated tracts; ≥3 and ≥7 give 46 and 15. Every row drops tracts below its threshold from the control pools rather than using them as controls, so the control pools are identical across rows and the dose is the only thing that changes.